

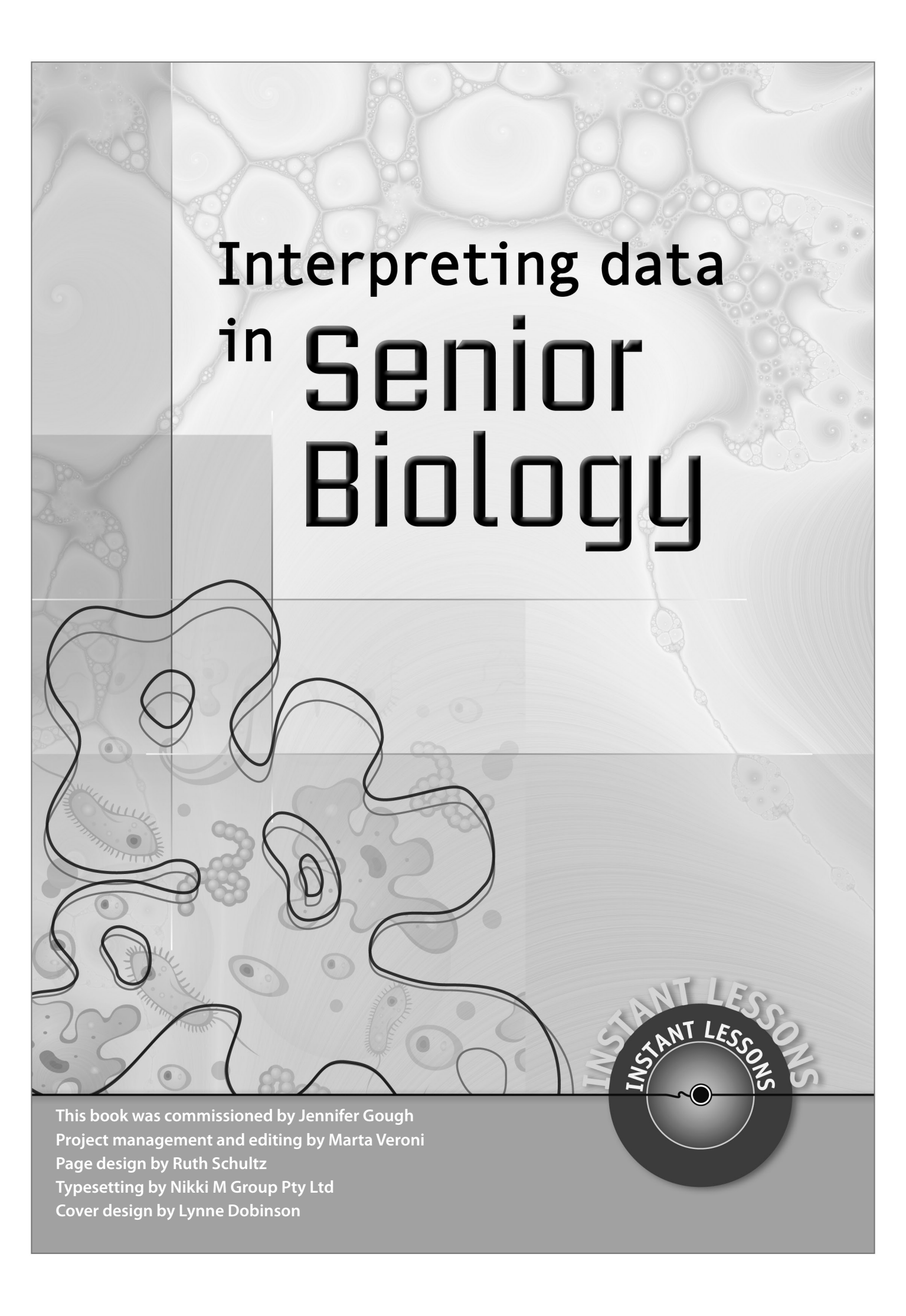
Interpreting data in **Senior Biology**

Peter Gribben

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The background of the cover features a grayscale microscopic image of cells. A prominent, thick, black wavy line, resembling a DNA helix or a protein structure, winds across the lower half of the cover. The title is centered in the upper half.

Interpreting data in **Senior Biology**



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INTRODUCTION

Edward Jenner (1749–1823) introduced the term *vaccination*. He discovered that by infecting people with cowpox (a mild disease) it made them immune to smallpox (a fatal disease). We now know that this immunisation is due to antibodies (immunoglobins) that are produced by the body in response to pathogens (bacteria and viruses) or toxins entering the bloodstream. Antibodies attack these invaders and then disappear once the infection is over. A much more rapid response is triggered if the same pathogens or toxins infect the body at a later time. When this happens, large numbers of antibodies are speedily produced and home in on the foreign bodies and destroy them.

Epidemiologists study the distribution, frequency, causes and controls of diseases in populations.

Aim

To improve our knowledge of how vaccines work and be able to understand the arguments for and against their use.

Data

Evidence box 1

There are several types of vaccines currently in use:

- Inactivated – These are cultivated viruses (or bacteria) that have been killed and cannot replicate (e.g influenza vaccine).
- Attenuated – These are active viruses or living bacteria but they have been weakened and can only reproduce very slowly. Booster injections are not usually needed for these.
- Virus-like particles – These particles are derived from a virus but are not infectious.

Evidence box 2

Since the introduction of vaccines, epidemiological studies have shown that the incidence of chicken pox, whooping cough, measles, mumps, polio, rubella and smallpox has declined to a fraction of their pre-vaccination rates.

Evidence box 3

The symptoms of chicken pox (varicella) are an itchy rash and red spot (pox) blisters all over the body. There are few complications and a very low death rate for healthy children, but the illness can be much more severe in adolescents and adults.

Evidence box 4

There are about 10 000 cases of whooping cough (pertussis) in Australia each year. Initial symptoms resemble the common cold. Coughing then develops, usually in bouts, followed by a deep gasp (or 'whoop'). It can be very serious for infants.

Evidence box 5

The symptoms of polio (poliomyelitis) vary from mild, flu-like symptoms to life-threatening paralysis. Australia was declared polio free in 2000 and the first case of polio in Australia since 1986 was reported in July 2007.

Evidence box 6

The initial symptoms of smallpox (variola major) are headache and fever followed by the development of small red spots which turn into blisters. If left untreated it can have a high mortality rate. The last recorded case of smallpox in Australia was in 1938.

Evidence box 7

The polio vaccine can in rare cases – about one in every 2.5 million doses – cause paralysis.

Evidence box 8

Some parents believe that the risks of routine childhood vaccinations are greater than the benefits. Their reasons for not immunising their children include the belief that vaccinations can cause chronic illnesses and 'overload' the immune system, vaccines are not always effective and nowadays the risks of vaccine-preventable diseases are low.

Evidence box 9

Epidemiologists insist that the risks for vaccination are low and that by not vaccinating them, parents are putting their children and other children at risk. Vaccination is not recommended for some children who are too young or whose immune systems are compromised.

Evidence box 10

For a disease to spread through a community, there must be both an infected individual to spread it and a susceptible individual to catch it. If a large number of people have been vaccinated, the disease will disappear from the community because there are not enough people to continue the catch-and-infect cycle. This is called 'herd immunity'. If immunisation rates drop to below 90%, it could provide the 'tipping point' to an outbreak of the disease.

Describing the data

- 1 Draw up a table showing the common and scientific name for each disease mentioned previously and list its symptoms.

Making links

- 2 Summarise the arguments for and against vaccination.

- 3 If Australia has been free of polio since 1986, what is a possible reason for a case occurring in 2007?

Questions

- 4 What is epidemiology?

- 5 When might a health professional advise against vaccinating an infant?

- 6 Give another name for antibodies. _____

- 7 Explain what is meant by the term 'herd immunity'.

INTRODUCTION

Taxonomy is a hierarchical system of naming and classifying organisms according to structural characteristics. The system we use was derived by Carolus Linnaeus (1707–1778) and gives every organism a unique name. The binomial system of nomenclature uses two words, e.g. *Homo sapiens* (modern humans). The first word is the genus; the second refers to the species. Note that the name is written in italics and the genus name has an initial capital letter.

Aim

To become familiar with how we classify animals, especially humans.

Data

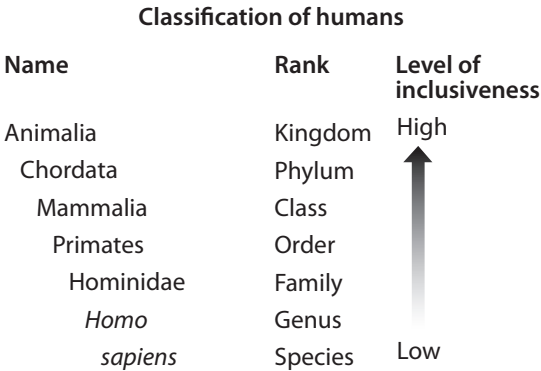


Figure 1

Treating the data

- 1 Write the distinguishing characteristics listed below next to the appropriate term (name) in Figure 1.
- animal bipedal human opposable thumb or big toe spinal cord suckles young wise

Interpreting the data

- 2 Figure 1 shows that as we go from ‘species’ to ‘kingdom’, the ‘level of inclusiveness’ goes from ‘low’ to ‘high’. What does this mean?

Questions

- 3 Why do biologists use the binomial system for naming organisms?
-
-
- 4 Outline the way organisms are named using the binomial system.
-
-
-
- 5 Mnemonics are memory aids. One used to help recall the classification hierarchy is **King Paul Cries Out For Good Soup**. Explain what each word signifies and then make up your own mnemonic.
-
-
-
-

Information

A cladogram is a branching diagram that represents the ways in which taxa are related and indicates possible evolutionary sequences. It shows the relationship of selected animals on the basis of their shared anatomical *features*.

Data
Figure 2 shows a cladogram of animals that all have a dorsal nerve cord (chordates). The hagfish is a primitive, jawless fish without paired fins and belongs to the class Agnatha.

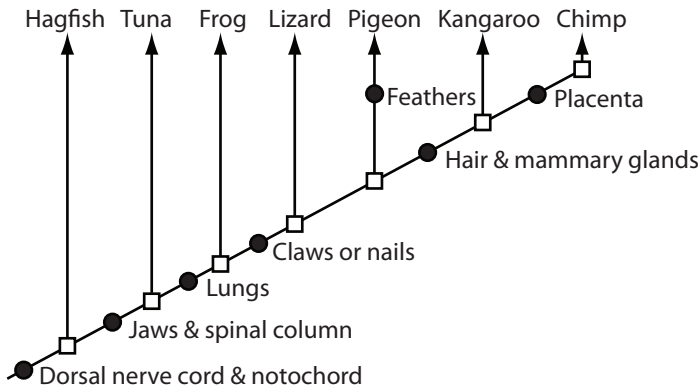


Figure 2

Treating the data

6 Complete the following table, showing the class and traits present in the animals. The first one has been done for you.

Animal	Class	Dorsal nerve cord	Jaws and spinal column	Lungs	Claws or nails	Feathers	Fur and mammary glands	Placenta
Chimp	Mammal	✓	✓	✓	✓		✓	✓
Frog								
Hagfish								
Kangaroo								
Lizard								
Pigeon								
Tuna								

Analysing the data

- 7 The traits that have been used to differentiate between the taxa in the cladogram are not the only ones that could be used. Next to each trait, write another that could be used. Remember that the trait has to apply to each animal that follows.
- 8 Where could humans be placed in this cladogram? Explain your answer.
- 9 Describe two traits that could be used to distinguish humans from the adjacent animal.
- 10 In the cladogram above (Figure 2), anatomical, i.e. structural, differences are used for classification. Outline two other types of differences that could be used.

INTRODUCTION

Angiosperms are flowering plants. Like all organisms, they need to reproduce and angiosperms do so sexually. Flowers are the sexual organs of these plants and they enable fertilisation to take place. Many, but not all, angiosperms have flowers that consist of both male and female parts.

Aim

To investigate pollination and how plants have evolved to ensure it.

Information

The male gametes (sex cells) are the pollen cells and the female gametes are the ovules. The male part of a flower is called the stamen and is made up of the filament (a thin stalk) and the anther, which is the source of pollen. The female part of the flower (the pistil) is located centrally. At the top is the stigma through which pollen can enter. The pollen then proceeds down a tube (the style) to the ovary where the ovules are located. Pollination is the transfer of pollen from the anther to the stigma and the subsequent fertilisation of the ovule. After pollination has occurred, the fertilised embryo develops into a seed, which can develop into a new plant under suitable conditions

Interpreting the data

- 1 Label the diagram in Figure 1, using the following terms:

- anther
- ovary
- stamen
- style
- filament
- pistil
- stigma

- 2 Draw lines to link the name of a flower part with its description.

Name	Description
Anther	Male structure
Filament	Female gamete
Flower	Has potential to form new plant
Ovary	Thin stalk
Ovule	Location of female gametes
Pistil	Location of male gametes
Pollen	Female structure
Seed	Tube linking stigma to ovary
Stamen	Sex organ of plant
Style	Male gamete

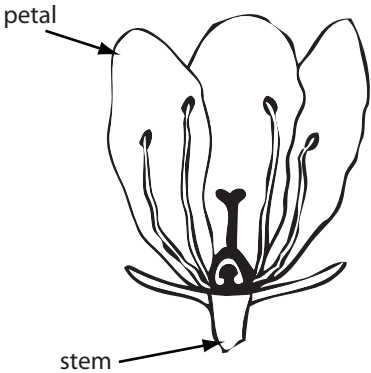


Figure 1

Information

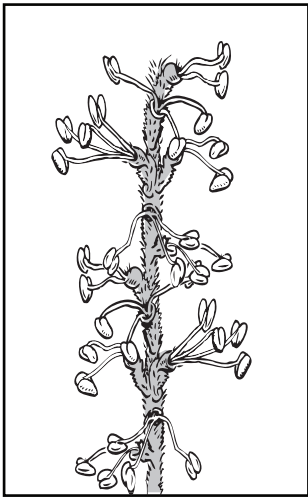
Pollen needs to be transferred to the stigma, and plants have evolved many ways to do this. External factors play a major role in this process. Such factors include wind and interactions with animals.

Plants that use wind as the pollinating vector produce large quantities of pollen and tend to have unobtrusive flowers with the anthers exposed. Wind is a good pollinator for species such as pine, which have dense populations and reach maturity at about the same time. When individuals of the same species are far apart or when there are many different species in the same area, the pollen is less likely to arrive at a plant of the same species.

In Australia, insects, birds and mammals participate in pollination. Less pollen is required than for wind-pollinated plants and it is a more efficient way of pollinating scattered or distant plants. However, plants have to attract animals, usually by providing food such as nectar, and this involves a cost to the plant.

Treating the data

3 Draw up a table that compares the advantages and disadvantages of wind pollination and animal pollination.



Plantago gaudichaudii

Figure 2



Banksia ericiflora

Figure 3

4 Figures 2 (*Plantago gaudichaudii*) and 3 (*Banksia ericifolia*) show two native Australian plants. State whether each is wind-pollinated or insect-pollinated. Give reasons for your answers.

5 Apart from pollen, what other parts of flowering plants can be moved by wind and animals?

Research questions

6 Apart from providing nectar, list other ways in which a plant could attract an animal.

7 Name an example of a plant that is pollinated by:

a an insect. _____

b a bird. _____

c a mammal. _____

Information

Genetic variation is important for plants. For this to occur, pollen from one plant must reach the stigma of a genetically different plant. There are two major ways in which plants avoid self-pollination. There can be male and female plants or, more commonly, when the pistil and stamen are both on the same flower they reach maturity at different times.

Seed set is the number of ovules that successfully develop into seeds. For successful seed set to occur, pollination must take place; but pollinated flowers don't always give successful seed sets.

8 Why is it important for plants of the same species to have genetic variation?

9 Give possible reasons why a pollinated flower might not give rise to seeds.

4 Global warming – possible impacts on the Pacific

INTRODUCTION

The Earth's climate has been constantly changing, and we are currently living in the middle of one such change. The vast majority of scientists acknowledge that global warming is taking place. Some causes blamed for global warming include:

- increased levels of greenhouse gases
- solar activity
- dust from volcanic eruptions
- part of a natural cycle.

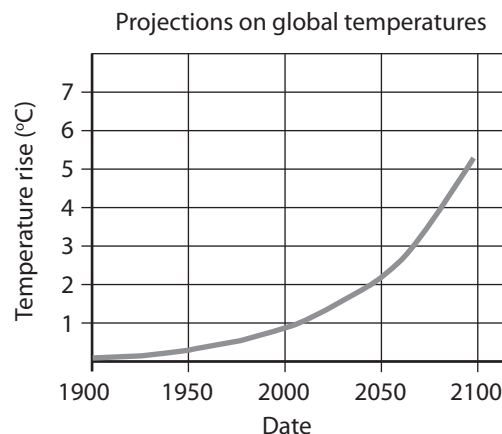
Although there is some disagreement, most scientists think that humans are the major contributors to this global warming due to the increase in emissions of carbon dioxide (a greenhouse gas) by the burning of fossil fuels.

Aim

To encourage critical thinking about global warming and its effects.

Information

The graph shows how the mean of air and ocean temperatures has varied since 1900. Figures from 2009 onwards have been projected using a variety of models. The data was gained mainly from European and North American sources.



Interpreting the data

The average air temperatures in Australia in 2009 ranged between a minimum of 16°C and a maximum of 28°C.

- 1 What were the average minimum and maximum air temperatures in Australia in 1900?

- 2 What are the average minimum and maximum air temperatures in Australia forecast to be in 2100?

- 3 What assumptions have been made when drawing the graph?

- 4 Discuss the assumptions mentioned.

- 5 Outline possible reasons why this data was sourced from Europe and North America rather than the Pacific region.

- 6 Discuss whether this data is applicable to the Pacific region.

4 – Global warming – possible impacts on the Pacific continued

Information

When global temperature rises, it causes a rise in sea level. When water is heated it expands. When ice on land melts, the water flows into the oceans. Many islands in the South Pacific are only a few metres above mean sea level and so are threatened by the rising waters. As well as land shrinkage, salt water is getting into water tables and affecting crop growth. There are few long-term records of sea levels in the South Pacific, but since the early 1990s more detailed measurements have been taking place. The table shows the rate of change in sea level between 1994 and 2007 of a typical island in the region. (Source: Australian Bureau of Meteorology)

Year	94	95	96	97	98	99	00	01	02	03	04	05	06	07
Rate of change (mm year ⁻¹)	+30	–25	–10	0	–25	–20	–10	0	+2	+3	+3	+4	+4	+3

Another, longer established institution had records showing that over the last 100 years, the sea level has been rising on average approximately 1.5 mm a year.

Interpreting the data

7 Comment on the trends in the sea-level data given in the table.

8 Discuss any differences between the two sets of data on sea levels.

9 Discuss the problems caused by rising sea levels.

Information

The diagram shows the profile of two South Pacific Islands. They are made of similar materials and have a mean height above sea level of four metres.



Interpreting the data

10 Which of the islands would be more affected by a rise in sea level? Explain your answer.

Information

Islands likely to be affected include Tuvalu, Kiribati, the Marshall Islands, Tokelau and the Carteret groups. Look at a detailed map of the islands of the South Pacific and find these islands.

Interpreting the data

11 Comment on the size and location of the affected islands.

12 What might people do, if the islands on which they live become uninhabitable as a result of a change in sea level?

INTRODUCTION

'Biodiversity is the variety of life: the different plants, animals and micro-organisms, their genes and the ecosystems of which they are a part. Australia is one of the most diverse countries on the planet. It is home to more than one million species of plants and animals, many of which are found nowhere else in the world.'

Australian Government, Department of the Environment, Water, Heritage and the Arts

Aim

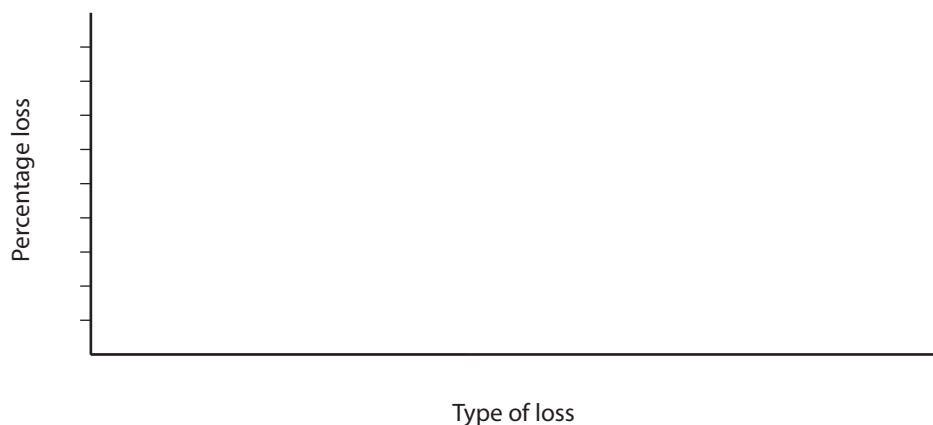
To examine human effects on diversity.

Data

Many scientists believe that the loss of native vegetation poses the single most significant threat to biodiversity. Europeans settled Tasmania in 1803 and since then it has lost 30% of its original native vegetation. The greatest losses are an estimated 66% of swamp forests, 47% of coastal heathland, 46% of dry forest and woodland, and 40% of grasslands. The percentage decrease is lowest in the vegetation types on the most infertile soils. In the period 1988–1994 an average of 10 500 ha were cleared annually. (Source: Australian Bureau of Statistics, 2002; Kirkpatrick and Jenkin, 1995.)

Treating the data

- 1 Using a column graph, plot the losses given in the data above.



- 2 Calculate the total area cleared between 1988 and 1994. Show your working.

Interpreting the data

- 3 Why might the loss of native vegetation be considered to be the greatest threat to biodiversity?

- 4 Give definitions for:

a diverse. _____

b species. _____

c biodiversity. _____

Information

There are many ways in which society tries to maintain biodiversity. One of these is to try to protect endangered species by individual attempts, legislation and institutions.

Data

The following are extracts from various newspaper reports.

Report 1
Under Section 18(3) of the national environment law – the *Environment Protection and Biodiversity Conservation Act 1999* – a person must not take an action that is likely to have a significant impact on a listed threatened species included in the endangered category.

Report 2
The Southern Tablelands Community Native Plant Society is looking for more volunteers to help with their projects of collecting and propagating native seeds, weeding and planting in the area. No special skills are needed.

Report 3
An environment department spokeswoman told our reporter that it was essential for anyone contemplating clearing an area or carrying out any activity that might impact on wildlife to check with the relevant environment department before starting work.

Report 4
A company was fined a total of \$270 000 for clearing trees that were designated as likely nesting habitat for the endangered south-eastern red-tailed black cockatoo. The magistrate told the defendants that the large financial penalty was imposed to show how seriously such conduct was regarded by the community and to deter possible future offenders.

Report 5
A local zoo has started a breeding program for red-tailed black cockatoos, which are becoming scarce. Once a large enough population has been established, individuals will be returned to the wild, hopefully to breed and increase numbers.

Report 6
A university researcher is recruiting amateur bird watchers to observe which birds use their bird baths. The information will be used to gain information about the abundance and location of many bird species.

Making links

5 Discuss how the strategies above aim to help biodiversity.

6 Outline other methods that could be used to support biodiversity.

INTRODUCTION

Ever since European settlement, plants and animals have been introduced into Australia. Some have been introduced on purpose, others have been introduced by accident.

Information

Salvinia molesta (a water fern) has become a major problem in some of Australia's bodies of water. Originally exported from Brazil for use in aquariums and garden ponds, it has escaped, probably as a result of people emptying aquariums and ponds into waterways. As *S. molesta* has no natural predators, it has flourished and become a major nuisance in some areas, especially Queensland.

This water fern grows best in slow-moving streams or still water ponds that have high nutrient levels and water temperatures of 20–30°C. Under optimal growth conditions *S. molesta* can double in volume every two to three days and completely cover an area of water. This can cause the following problems:

- Water quality is decreased.
- It provides cover for mosquitoes.
- Livestock have difficulty accessing drinking water.
- Livestock can become entangled and drown.
- The movement of aquatic animals is restricted.
- Recreational activities can be prevented.

Means of control of *S. molesta*, which has being declared a pest by government authorities, include:

- mechanical means – removal by hand or machine, a labour-intensive method that is only suitable for small areas or restricted growth
- herbicides – these cause problems with use of water for drinking
- biological controls – insect predators. These must not cause additional problems.

Aim

To encourage students to consider the problems involved with introduced flora and the means to control these.

Questions

- 1 Give two reasons why *S. molesta* can cover large areas of water so quickly.

- 2 Name two recreational activities on bodies of water.

Treating the data

- 3 Draw up a table that lists the advantages and disadvantages of each means of control of *S. molesta*. Try to include one possible advantage/disadvantage not already mentioned.

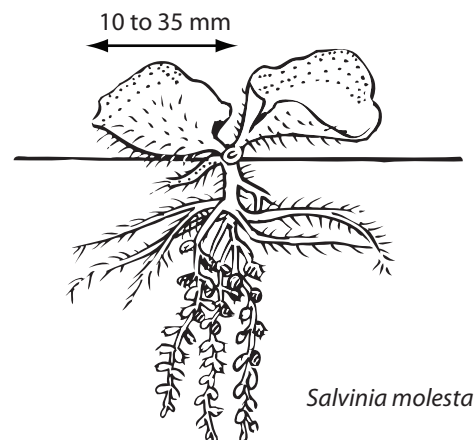


Figure 1

Information

One way to measure the growth of *S. molesta* and other weeds is to take aerial photos and use a grid. The number of squares covered is measured and the percentage cover is calculated. To simplify calculations, 'rounding up/down' is used. If a square is less than half full, it is counted as zero and if it is more than half full it is counted as one.

Treating the data

4 Calculate the percentage of surface covered by *S. molesta* in Figure 2. Show your working.

5 Comment on the accuracy of your result and how it could be improved.

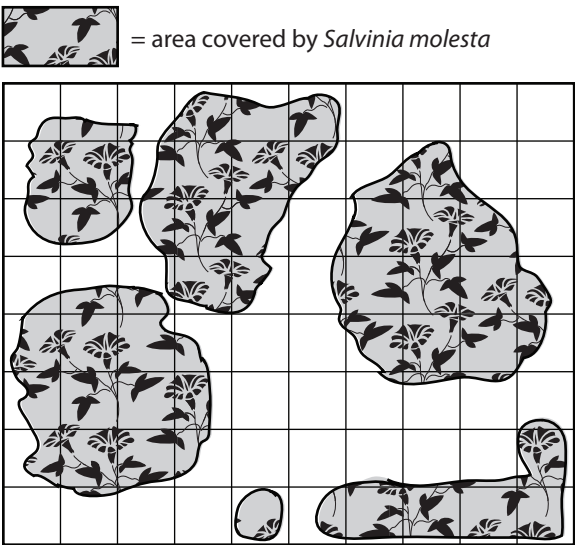


Figure 2

Information

A tiny South American weevil, *Cyrtobagous salviniae*, has been used successfully to control *S. molesta*. This insect prefers air temperatures between 27 and 35°C and flourishes when the plants have high levels of nitrogen present. The larvae feed on new growth buds and tunnel into the plant, causing internal damage. Adult weevils also prefer to eat plant buds and the combined attack can kill the plants. Figure 3 shows the percentage cover of *S. molesta* on a body of water over a period of time.

Interpreting the data

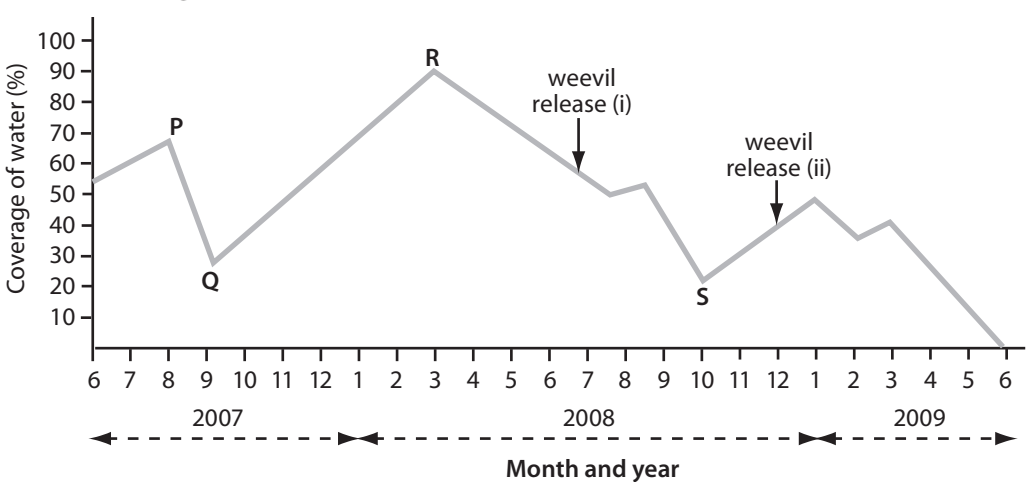


Figure 3

6 Give a possible reason for the decline in the amount of *S. molesta* between P and Q.

7 Give a possible reason for the increase in the amount of *S. molesta* between Q and R.

8 The initial weevil release was judged to be a failure because the percentage cover just followed the previous seasonal changes. Why did the second weevil release succeed?

INTRODUCTION

Exogenesis (literally 'outside origin') is a theory that life began elsewhere in the universe and travelled to Earth. Geogenesis proposes that life arose independently on Earth.

Panspermia, from the Greek *pan*, 'all', and *sperma*, 'seed', is a theory that states that the universe is full of spores/seeds that germinate when they find a favourable environment. The theory is not new, but has been revived at intervals since the 18th century. Many reputable scientists including Jöns Jacob Berzelius (1779–1848), Lord Kelvin (1824–1907), Hermann von Helmholtz (1879–1927) and Svante Arrhenius (1859–1927) have proposed versions of panspermia. Many in the scientific community believe that it is highly probable that life exists elsewhere in the universe because of the immense numbers of potential sites. Until we can survey our galaxy in detail for signs of life, the panspermia hypothesis will remain difficult to support or disprove.

Aim

To analyse evidence in support of the theory that life on Earth occurred as a result of extraterrestrial life forms landing on Earth.

Information

Box 1

Sir Fred Hoyle (1915–2001) and Chandra Wickramasinghe (1939–) argued that life forms have continued to enter the Earth's atmosphere, and that these are responsible for outbreaks of new diseases and the genetic novelty necessary for macroevolution. Using spectroscopic analysis of light from distant stars, they claim to have found evidence that traces of life exist in the dust between Earth and these stars.

Box 2

Arrhenius proposed that radiation pressure from sunlight (electromagnetic radiation) could propel spores through space. Kelvin thought that micro-organisms could travel in rocks (lithopanspermia), which would provide protection from radiation.

Box 3

The majority of scientists think that interstellar panspermia, i.e. travelling between stars, is unlikely, given the harsh conditions, such as a vacuum, extremes of temperature and cosmic radiation, in space.

Box 4

In 1976 the Viking Mars lander carried out biological experiments which gave results that were initially indicative of life being present in Mars soil. However, no organic molecules were found in these soils and similar results were obtained from heated controls. It is likely that the results were due to abiotic chemical reactions rather than the metabolism of living organisms.

Box 5

Interplanetary exchange of materials has occurred on many occasions. In 1984 a 4.5 billion-year-old rock, labelled meteorite ALH84001, was discovered in Antarctica. It is thought to have originated from Mars and to contain fossil evidence that primitive life may have existed on Mars more than 3.6 billion years ago.

Box 6

Evidence that meteorites carry extraterrestrial life forms is not clear cut. Some claims have been proven to be unfounded as a result of terrestrial contamination or misinterpretation. Others are currently hotly disputed.

Box 7

Extremophiles are durable micro-organisms that can survive extraordinary conditions on Earth. Existence of these supports the view that simple life forms are capable of surviving long interstellar journeys.

Interpreting the data

1 Which information boxes support the theory of panspermia?

2 Which information boxes contradict the theory of panspermia?

3 Discuss the statement: 'Panspermia does not remove the need for life to originate somewhere.'

4 Give definitions for:

a abiotic. _____

b interstellar. _____

c control. _____

d metabolism. _____

e extraterrestrial. _____

f extremophile. _____

g contamination. _____

Research

Write your answers to questions 5 and 6 as information boxes.

5 Research two more pieces of information that support the theory of panspermia.

6 Research two more pieces of information that refute the theory of panspermia.

7 Outline the theory of geogenesis.

Making a concept map

8 Use the data in the information boxes and your research to make a concept map of panspermia. It is recommended that you work in groups for this activity.

a In the centre of a sheet of paper write 'Panspermia'.

b Cut out all of your information boxes including the ones you wrote.

c Group related boxes together around the term 'Panspermia'.

d Draw arrows from 'Panspermia' to the boxes and between linked boxes.

e Label the arrows to describe why the boxes are linked.

f Compare your concept map with those of other groups.

INTRODUCTION

Scientists' thoughts on the platypus have changed over the past 200 years as new information and technologies became available. Early descriptions of the platypus seemed far fetched to those who had not seen it. When the first preserved specimen reached England in 1798, it was initially thought to be a hoax. It seemed that someone had stuck together the bill of a duck, the skin of a mole and the tail of a beaver. Closer examination and the arrival of further specimens proved that the platypus did indeed exist.

Aim

To increase students' appreciation of how scientists gain knowledge of a species.

Information

The original name for the platypus was *Platypus anatinus*, which means 'flat-footed and duck-like'. This was later changed to *Ornithorhynchus anatinus*. The platypus is a member of the order Monotremata. The only other monotremes are echidnas. The order Monotremata means 'one hole' – there is a single opening serving the digestive, excretory and reproductive systems.

The platypus confused early taxonomists because:

- it was an endotherm (warm blooded)
- its body temperature is lower than that of most mammals
- it has fur
- it has a tail like a beaver
- it has feet like an otter
- it has mammary glands
- it lays eggs
- its bill is similar to that of a duck
- the male produces venom in spurs on its legs.

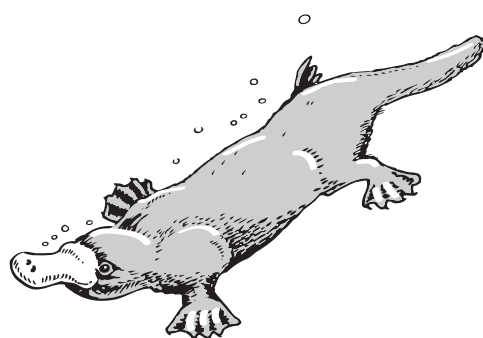


Figure 1

Interpreting the data

- 1 If the drawing of the platypus (Figure 1) is at a scale of 1:5, what is its actual size?

- 2 Complete the classification table.

Platypus classification	
Kingdom	Animalia
Phylum	Chordata
Subphylum	Vertebrata
Class	Mammalia
Subclass	Prototheria
Order	
Family	Ornithorhynchidae
Genus	
Species	

- 3 Discuss which of the above characteristics of the platypus are consistent with it being a mammal and which are more likely to be associated with other classes.
-
-
-
-
-

Question

- 4 What is a taxonomist?
-

Information

Phylogeny is the sequence of events involved in the evolutionary development of a species or taxonomic group of organisms. Figure 2 shows the phylogeny of mammals. One major difference between monotremes, marsupials and placental (eutherian) mammals is the way in which their offspring are born. Monotremes lay eggs, marsupials give birth to immature young and carry their young in pouches and placental mammals have a placenta – the organ that connects the developing foetus to the uterine wall so that nutrient uptake, waste elimination and gas exchange can occur – and give birth to developed young. Monotremes are found only in Australia and New Guinea. Marsupials are found in Australasia and South America, and placentals are found on all continents.

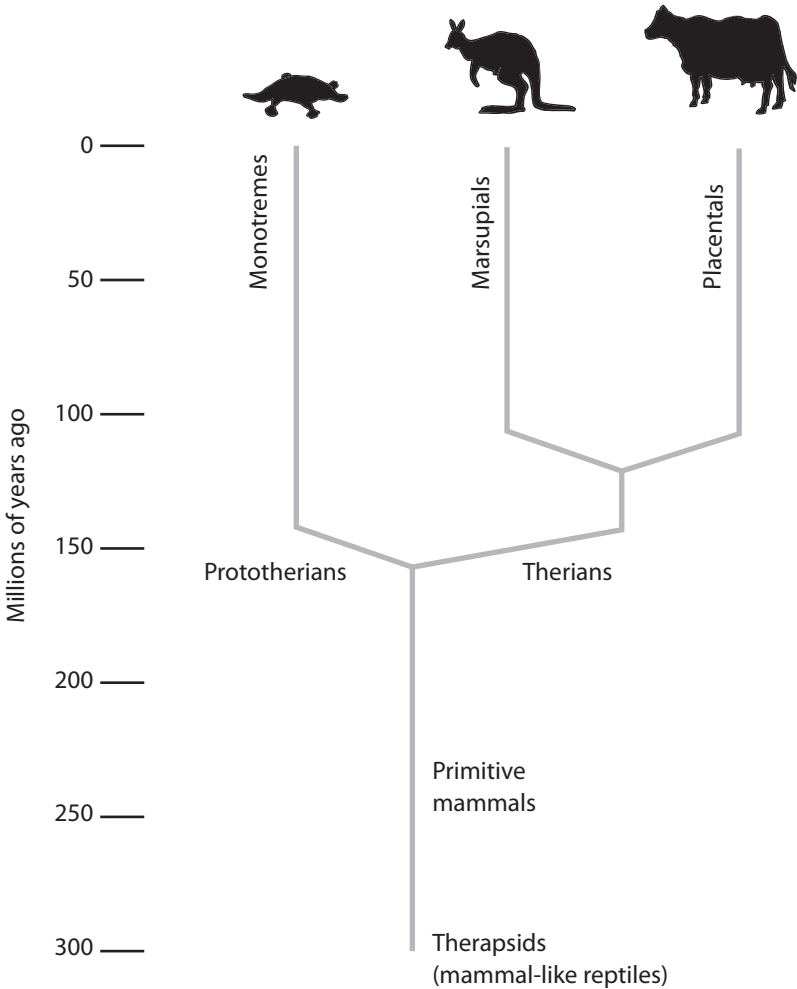


Figure 2

Interpreting the data

- 5 Approximately how long did it take primitive mammals to evolve from therapsids?
- 6 How many millions of years ago (mya) did marsupials and placental animals start to evolve from therians?
- 7 At one time scientists thought that the platypus was a ‘missing link’ or transition between reptiles and mammals. Does Figure 2 support or contradict this view? Explain your answer.
- 8 Complete the table.

Mammal	Lay eggs	Have a pouch	Suckle young	Location of species	Endothermic or ectothermic	Covered in skin and fur
Monotremes						
Marsupials						
Placentals						

Information

Scientists have gained knowledge on the platypus by observing it in the wild, breeding it in captivity and studying its DNA and fossils. Fossils studies have shown that a type of platypus existed in South America about 60 mya.

Question

- 9 This fossil evidence supports which theory about continents?

Research

- 10 Read the following passage and define the words in *italics*.

The platypus has a thick *pelage* to maintain its temperature when feeding in very cold waters. It is an *obligate* aquatic feeder and is usually *nocturnal* in its behaviour. It has a unique electrosensory system in its bill to help it locate prey. The platypus has a *gestation* period of about a month.
- a pelage
- b obligate
- c nocturnal
- d gestation

INTRODUCTION

'Biological grading' is one way to assess water quality. This method is based on the presence of aquatic macro-invertebrates in a body of water. Macro-invertebrates are small animals that can be seen with the naked eye. They include insects such as dragonfly nymphs, mosquito larvae, water-striders and mayflies, as well as snails, shrimps and worms. Macro-invertebrates are extremely useful for biological assessment because they:

- are found in virtually all fresh waters
- are easy to collect
- are sensitive to changes in their ecosystem
- tend to stay within specific areas.

Aim

To analyse data showing the change in water quality over time.

Data

The inland water systems of three adjacent states were assessed by biological grading and the results shown in Figure 1. In State C, the total length of waterways analysed increased by more than half between 2004 and 2009.

Categories were good and moderate.

Processing the data

- 1 Write the data in table form.

Interpreting the data

- 2 Comment on the trends shown.

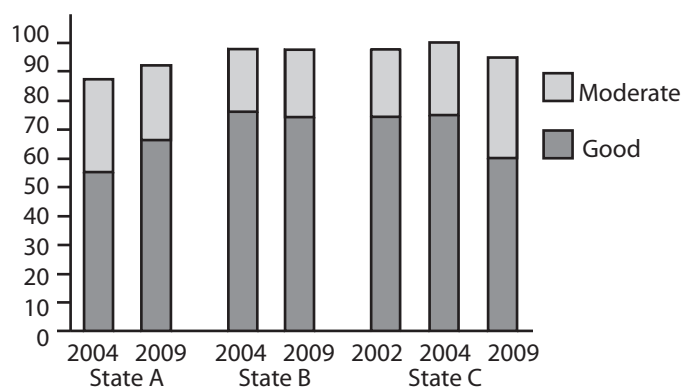


Figure 1

- 3 State C invested heavily in antipollution measures between 2000 and 2009 to improve water quality. Give possible reasons for the apparent overall decrease in its water quality.

Information

There are many methods that can be used for biological grading but they have the following in common:

- Organisms are graded according to their sensitivity to pollution.
- The number of organisms in an area is counted. Some systems count the total numbers of organisms, others count the number of different species or families.

Aim

To work out the water quality of a body of water using biotic values.

Data

This table shows a typical sample.

Animal	Biotic value	Number found	Animal	Biotic value	Number found
Stonefly nymph	10	13	Damselfly nymph	6	3
Mayfly nymph	10	36	Water strider	6	4
Dobsonfly larva	10		Freshwater shrimp	4	
Adderfly larva	10		Midge Larva	4	21
Caddisfly larva	8		Crayfish	4	
Beetle larva	8	15	Snails	2	22
Crane fly larva	8	4	Mosquito larva	1	
Dragonfly larva	8	17	Leech	1	
Freshwater mussel	6		Aquatic worm	0	

Treating the data

The higher the biotic value, the more sensitive an organism is to pollution. Using this scheme, the biotic index value is found by multiplying the numbers of each animal found by its biotic value, adding the numbers and dividing the total by 10.

- 4 Calculate the value of the biotic index for this body of water, showing working.

Water quality	Biotic index value
Excellent	Over 75
Good	60–74
Fair	40–59
Poor	Below 40

- 5 What is the water quality of this body of water?

Interpreting the data

- 6 Describe factors, other than pollution, which could affect the numbers of organisms counted.

Questions requiring research or interpretation

- 7 What is the major difference between micro-invertebrates and macro-invertebrates?
- 8 Apart from the presence or absence of organisms, what other measurements can be made to measure water quality?

INTRODUCTION

Vitamins are essential for the growth and function of cells and for the body’s metabolism. Vitamins are complex molecules that help reactions occur. They are only needed in minute amounts, typically micrograms (µg) or milligrams (mg). Some scientists are concerned that people may be taking an excess of vitamins, hoping to keep healthy, but this can cause a health risk.

Aim

To increase understanding of suitable levels of vitamin intake.

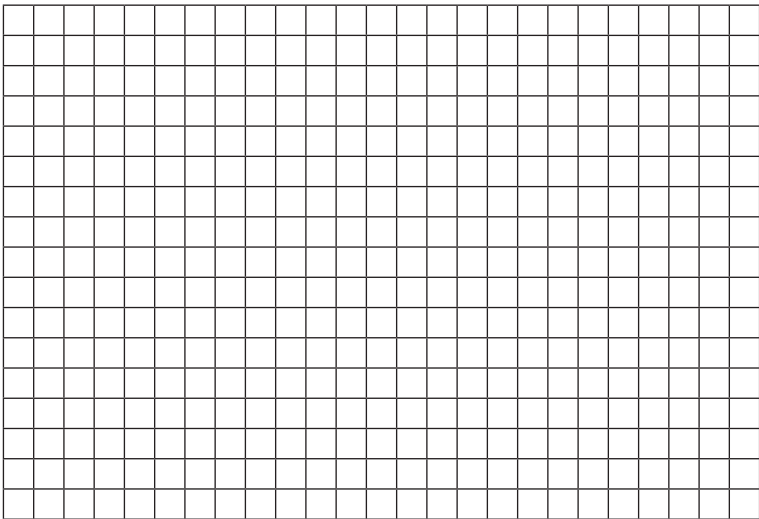
Data

Vitamin	*RDI (mg) for adults		Helps with	Soluble in
	Minimum	Maximum		
A	0.7	3.0	Bones, skin and vision	Fat
B2	1.5	15	Energy metabolism	Water
C	82	2000	Removing oxidants (antioxidant)	Water
D	0.01	0.05	Immune system and bone formation	Fat
E	15	1000	Formation of red blood cells	Fat
K	0.008	0.01	Blood clotting	Fat

*RDI = recommended daily intake

Treating the data

I Draw a column graph showing the minimum and maximum levels (y-axis) for vitamins A, B2, D and K (x-axis).



Interpreting the data

- 2 Why were you not asked to include vitamins C and E in your graph?
- 3 Which vitamin has the greatest percentage range between minimum and maximum intakes?

Information

Any water-soluble vitamins taken in by the body but not used are excreted in urine, whereas fat-soluble vitamins are stored in the body. This is important if people’s intake of vitamins is irregular or excessive.

Question

- 4 Draw up a table showing the advantages and disadvantages of vitamins that are water soluble and those that are fat soluble. Consider what might happen if a vitamin was taken in excess.

Information

The label below is of a typical vitamin and mineral supplement.



multi-vitamin supplement

Dosage: take one tablet, two times daily

Each tablet contains:	
Vitamin A	0.5 mg
Vitamin B2	0.6 mg
Vitamin C	110 mg
Vitamin D	0.0 mg
Vitamin E	800 mg
Calcium	90 mg
Iron	8 mg
Magnesium	32 mg
Potassium	13 mg
Zinc	2 mg

This product is designed as a supplement and may help in maintaining general well being, especially when intake of vitamins and minerals from diet is insufficient.

This supplement will not replace a balanced diet.

Do not take greater than recommended dosage.

Treating the data

- 5 Draw a table that shows the percentage of the minimum and maximum RDI of each vitamin that the supplement supplies.

Interpreting the data

- 6 Comment on the amounts of vitamins supplied by the product.

- 7 Would a person using this supplement necessarily be unhealthy if the supplement did not supply 100% of required nutrients? Explain your answer.

INTRODUCTION

The ears are the organs of hearing and balance. The outer ear is made up of the pinna and the ear canal. The middle ear contains the eardrum (tympanic membrane) and the ossicles ('hammer','anvil' and 'stirrup'). The inner ear contains the round window, the oval window, the cochlea, the semicircular canals and the auditory (cochlear) nerve.

Aim

To become familiar with the names, structure and functions of various parts of the ear.

Data

The boxes below contain information about the parts of the ear.

Interpreting the data

I Cut out Figure 1 and paste it on an A4 sheet of paper. Label the outer ear, the middle ear and the inner ear.

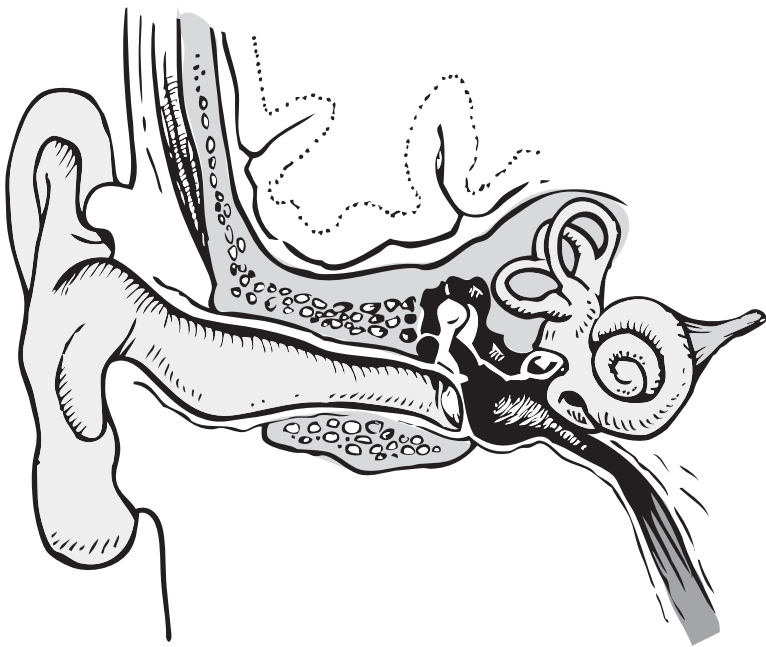


Figure 1

2 Cut out the boxes and use them to identify the parts of the ear, their structure and their role.

Shell of skin and cartilage	Flexible tissue joining outer and middle ear	Connects middle ear and pharynx	Connects ossicles to inner ear
Pinna	Oval window	Three looped cavities	Ear ossicles
Spiralling tubular cavity	Oval hole in skull	Auditory nerve	Eustachian tube
Controls balance	Ear canal	Short tube	Equalises air pressure
Round hole in skull	Chain of three tiny bones	Cochlea	Round window
Collects sound	Tympanic membrane	Moves ossicles	Semicircular canals
Changes sound to electrical signal	Amplify movements of eardrum	Allows fluid in the cochlea to move	Connects external ear to eardrum
	Takes electrical signal to the brain	Collection of nerve fibres	

INTRODUCTION

Information obtained from practical procedures can be first-hand or second-hand.

First-hand investigations are those that you carry out yourself, e.g. practicals carried out at school or at home. Secondary sources refer to information you may have obtained from elsewhere, e.g. from text books, encyclopaedias or the internet. For each investigation it is important to state a hypothesis, identify the variables involved and state that the investigation is valid, i.e. if it tests what it says it is testing.

Aim

To enable students to form hypotheses and identify the different types of variables.

Information

A hypothesis is a statement that can be tested by scientific method. It is often described as an educated guess based on observations. Factors that can be changed during an experiment are called *variables*. Usually only one variable is changed at a time. There are three major types of variables:

- independent
- dependent
- controlled.

The independent variable is the major part of the hypothesis (belief) that you are testing. It is the condition that you change to see what happens. The dependent variable is the thing that changes when the independent variable is changed. A controlled variable is one that is kept the same in each experiment.

Data

A student had problems with whitefly in her greenhouse in which she was trying to grow tomatoes. She found an article in a gardening magazine which stated: 'Whiteflies are strongly attracted to the colour yellow. A cheap way of trapping them is to cut out strips of yellow plastic, coat these in clean engine oil and hang them near the affected plants. The insects become stuck on the oil.' The student set up the devices as shown in Figure 1, using strips of yellow, blue, green and black plastic and measured the number of whiteflies trapped over a 48 hour period on each strip.

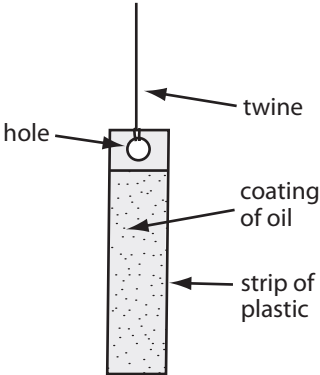


Figure 1

Describing the data

1 Write a hypothesis and identify the variables for the previous investigation.

Hypothesis	Variables		
	Independent	Dependent	Controlled

Interpreting the data

2 Outline any further steps the student could carry out to ensure that any conclusions she made were valid.

Information

Rennin is an ingredient of junket tablets. If mixed with milk, it causes milk to solidify. As it is an enzyme, rennin works best at a specific temperature.

Data

A student was set the task to find the temperature at which junket tablets worked quickest. Part of his report is shown below.

Method

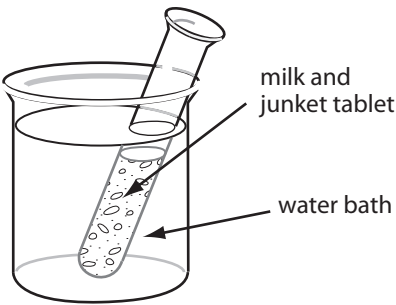
Milk in a test tube was placed in a water bath and the temperature of the milk recorded. One-quarter of a junket tablet was crushed and added to the milk. The time taken for the mixture to solidify was measured. This was repeated over a range of temperatures.

Results

Temp. (°C)	10	20	30	40	50
Time (min)	13.7	8.9	6.6	4.5	7.8

Conclusion

The optimum temperature at which junket tablets set milk is 40°C.



Describing the data

3 Write a hypothesis and identify the variables for the previous investigation.

Hypothesis	Variables		
	Independent	Dependent	Controlled

Interpreting the data

4 Outline any further steps the student could carry out to ensure that the conclusion he made was valid.

Making links

5 Outline how you could test the following hypothesis. Identify the variables in your test.

The action of junket tablets is affected by the pH of the milk to which they are added.

Hypothesis	Variables		
	Independent	Dependent	Controlled

INTRODUCTION

Crocodiles have been on Earth in their present form, or close to it, for millions of years. Although differences exist between species, crocodiles tend to have long jaws, a protective armour, a streamlined body and a long tail. They play an important part in freshwater and estuarine ecologies.

Aim

To increase knowledge of the roles crocodiles play in their environment and how well they are adapted to their environment.

Information

Two species of crocodiles exist in Australia: the saltwater (estuarine) crocodile, *Crocodylus porosus*, and the freshwater crocodile, *Crocodylus johnstoni*. They are often referred to respectively as 'salties' and 'freshies'. Despite the names, saltwater crocodiles are also commonly found in fresh water and freshwater crocodiles, although they prefer fresh water, can sometimes be found in slightly saline water, if the more aggressive saltwater crocodiles are not present. Saltwater crocodiles grow to much larger sizes than freshwater crocodiles and have broader snouts. Both species are almost exclusively carnivorous, but only saltwater crocodiles can catch large prey such as cattle. Freshwater crocodiles breed in the Northern Australian dry season (July to September), whereas saltwater crocodiles breed in the wet season.

Treating the information

- 1 Draw up a table comparing the characteristics of the two types of Australian crocodiles.

Question

- 2 Why might it be reassuring to see some freshwater crocodiles in a body of water?

Data

Figure 1 shows (to scale) the relative sizes of two different types of adult crocodile compared to a human.

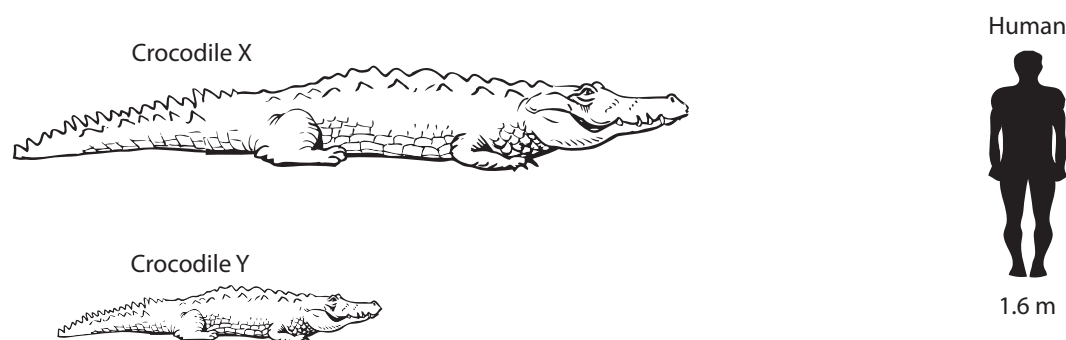


Figure 1

Interpreting the data

- 3 Calculate the lengths of the two crocodiles. Show working.
- 4 Identify the species to which the crocodiles belong. Explain your answer.

Data

Crocodiles bask on mud banks and leave tracks, which help scientists to gain information on crocodiles in the area. These tracks indicate a crocodile's size, weight and behaviour.

Figure 2 shows two such imprints.



Identified crocodile



Unidentified crocodile

Figure 2

Interpreting the data

- 5** The identified crocodile was known to be 2.9 m in length. Calculate the probable length of the unidentified crocodile. State any assumptions you made.

- 6** Explain how tracks could give information on the mass of a crocodile.

Information

Crocodiles are reptiles and are closely related to dinosaurs. They have changed little since dinosaurs disappeared more than 65 million years ago. The reason for the lack of change is that crocodiles are very well adapted to their environment. The following list includes some of the characteristics of crocodiles:

- | | |
|--|--|
| a are ectotherms ('cold-blooded') | b have a four-chambered heart |
| c need warm wet conditions | d can stay underwater for hours at a stretch |
| e lay eggs | f have a dry scaly skin |
| g have body armour | h can strike very quickly at prey |
| i can eat a wide range of food | j can live in fresh and salty water |
| k can swim in water and walk on land | l live for many years |
| m have a keen sense of smell | n can detect vibrations in water |
| o have exceedingly strong jaws | p have a high mortality rate among hatchlings |
| q have their sex determined by the temperature of the hatching eggs | |
| r have eyes that are adapted to their lifestyle and environment | |
| s can survive for long periods between meals | |
| t use less energy than a mammal of equivalent weight | |

Questions

- 7** Which of the above statements apply to reptiles in general?
- 8** Which of the above statements apply to crocodiles but not to other reptiles?
- 9** Which of the above statements have made crocodiles so successful at surviving?

INTRODUCTION

Crops can be defined as produce that appears in a particular season or place. Humans have been growing crops for at least 10 000 years. Cereal crops such as wheat, barley, corn and maize are hugely important for the Australian economy, amounting to billions of dollars in exports. Farmers use many methods to maximise crop yields.

Aim

To analyse data relating to crop growth and gain an appreciation of some of the factors involved.

Data

A seed company trialled two different types of seed (varieties A and B), planting at different times and in various areas to investigate which seed and conditions would give the best yield. They used a low rainfall zone (less than 325 mm year⁻¹), a medium rainfall zone (325–400 mm year⁻¹) and a high rainfall zone (over 400 mm year⁻¹). The following table shows the results.

Rainfall zone/ Sowing time	Yield (kg ha ⁻¹)	
	Variety A	Variety B
Low rainfall		
Early	3400	4300
Middle	3100	3700
Late	2100	2200
Medium rainfall		
Early	4400	5130
Middle	4300	4200
Late	3000	2600
High rainfall		
Early	4600	4900
Middle	4350	4050
Late	3400	2900

Processing the data

- I
- Use the information in the table to draw a graph.



Analysing the data

- 2
- Summarise the information given.

Data

Plants use light energy and the raw materials carbon dioxide (CO₂) and water (H₂O) to form carbohydrate (glucose, C₆H₁₂O₆) and oxygen (O₂). A simplified equation for the process of photosynthesis is:



A commercial grower wanted to increase the yield of greenhouse-grown tomatoes and investigated if increasing the concentration of carbon dioxide would have an effect. Figure 1 shows the results of the trial. The first reading is at normal atmospheric concentrations of carbon dioxide.

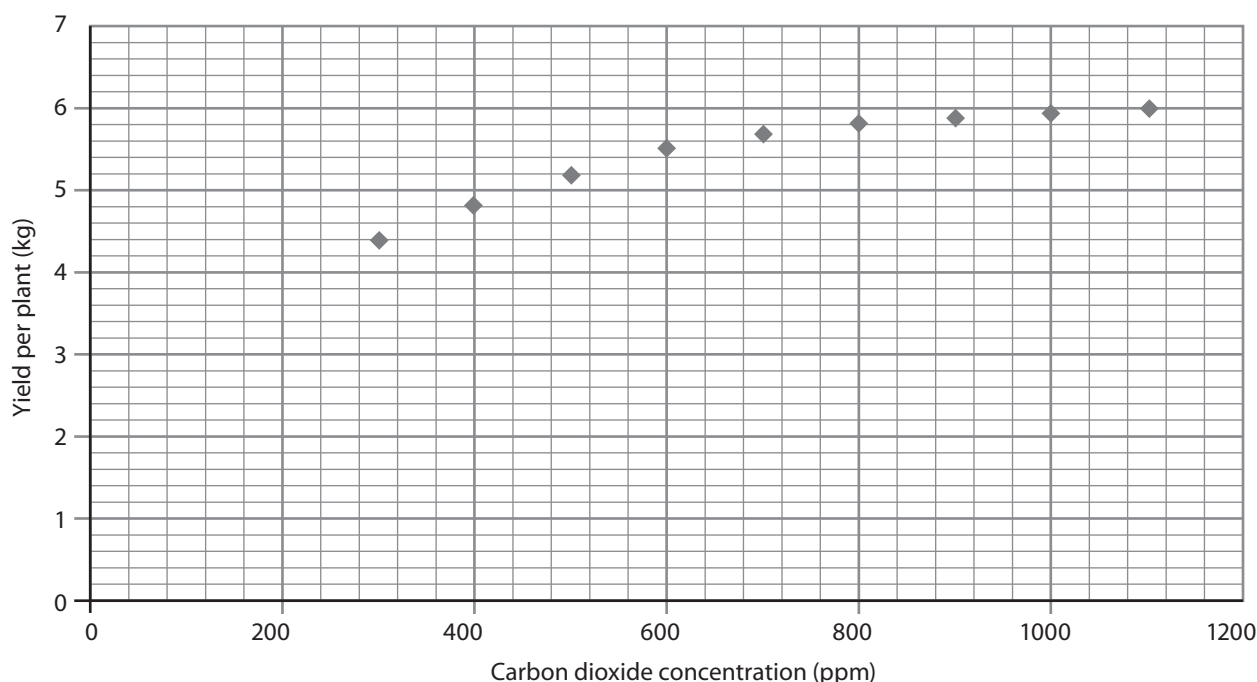


Figure 1

Interpreting the data

- 3 What is the normal atmospheric concentration of carbon dioxide? _____
- 4 Calculate the percentage increase in yield of tomatoes at the following carbon dioxide concentrations. Show working.
 - a 500 ppm _____
 - b 800 ppm _____
 - c 1100 ppm _____

Making links

- 5 Give possible reasons why increasing the carbon dioxide level does not cause proportional increases in yield of tomato plants.

Questions

- 6 Write a word equation and then a balanced formula equation for the process of photosynthesis.

INTRODUCTION

A chronic disease is one that has been present for a long time, e.g. 3 months or more. Chronic heart failure (CHF) is a condition in which the heart is no longer able to pump blood efficiently through the body. Although there have been advances in treating the disease, it remains a major public health problem in industrialised nations. This disease reduces quality of life considerably and has a high mortality rate. Treatments include diet, exercise, pharmaceuticals and surgery including heart transplants.

Aim

To investigate and analyse some causes and treatments of CHF.

Data

Evidence box 1

Risk factors include:

- alcohol abuse
- diabetes
- disease of the heart muscle (cardiomyopathy)
- heart valve disease
- high blood pressure
- obesity
- previous heart attack (myocardial infarction)
- smoking.

Evidence box 2

Symptoms include:

- shortness of breath
- impaired kidney function
- impaired liver function
- congestion of the lungs
- intestines become less efficient
- eventual death.

Evidence box 3

Lifestyle modifications include:

- reduced total intake of food, if obese
- reduced intake of sodium and saturated animal fats
- moderate exercise.

These changes are more important in preventing CHF than treating it, but can help in most cases.

Evidence box 4

Pharmaceuticals include:

- ACE (angiotensin-converting enzyme) inhibitors, which slow the loss of heart pumping ability
- diuretics to reduce excess liquid in body
- digitalis improves strength of heart's contractions.

Drugs can both improve the quality of life and extend a patient's life.

Evidence box 5

Surgical techniques include:

- coronary bypass to detour around blocked blood vessel
- myocardial revascularisation to provide new pathways through heart muscle
- pacemaker, which provides regular stimulus to the heart
- heart transplant
- installation of artificial heart
- artificial heart valves.

Surgical techniques are effective in many cases. There are insufficient donors for heart transplants and artificial hearts are regarded as 'stop gaps'.

Evidence box 6

Statistics show that:

- there are 1.2 patients per 1000 population
- more men than women are affected
- mortality increases with age
- total mortality has decreased over the past 20 years
- the CHF mortality ratio is significantly higher in Aborigines and Torres Strait Islanders.

Describing the data

- 1 List the risk factors for CHF that are most likely to be related to a person's lifestyle.

- 2 List three symptoms of CHF.

- 3 State two lifestyle modifications not mentioned in evidence box 3 that could alleviate or help prevent CHF.

4 What do ACE inhibitors do to help in CHF?

5 How can digitalis help in CHF?

6 What is a coronary bypass?

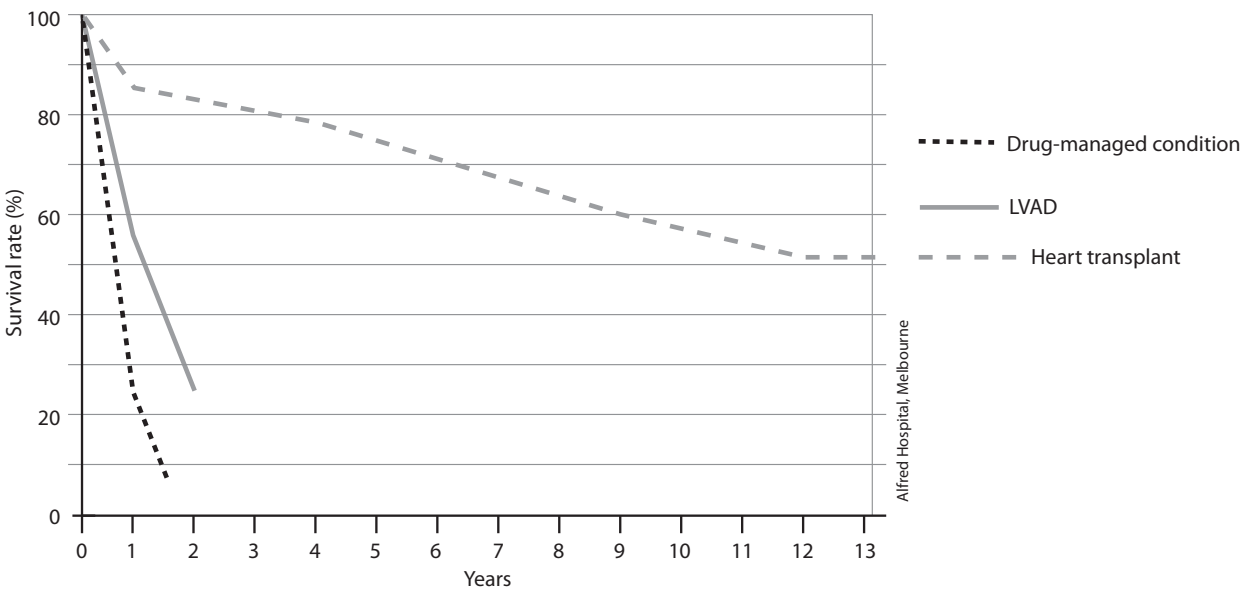
Interpreting the data

7 If the current population of Australia is 21 500 000, calculate the number of people with CHF. Show your working.

8 Give possible reasons why the total mortality due to CHF has decreased in Australia over the past 20 years.

Data

Left ventricular assist devices (LVADs) help the heart to pump rather than replacing it. Left ventricular assist devices and total artificial hearts (TAHs) can keep patients alive until a donor heart becomes available.



Source: Rachel Nowak, *New Scientist*, 31 October 2004.

Interpreting the data

9 What is the difference between a LVAD and a TAH?

10 What is the 1 year survival rate of a patient using drugs?

11 Although heart transplants have the highest survival rate, only a small percentage of CHF sufferers get a transplant. Give possible reasons for this.

INTRODUCTION

The word 'megafauna' literally means 'large animals', but it usually refers to extinct animals that were much larger than their counterparts of today. Australian megafauna existed in large numbers until about 50 000 years ago, though dates for their extinction vary considerably. Two major theories exist for their extinction: climatic change and human intervention.

Remember to distinguish between *extinct* species that no longer exist and *extant* species that are still in existence.

Aim

To use data to evaluate possible explanations for the extinction of Australian megafauna.

Data

Evidence box 1

Megafauna flourished and then became extinct in the Pleistocene epoch.

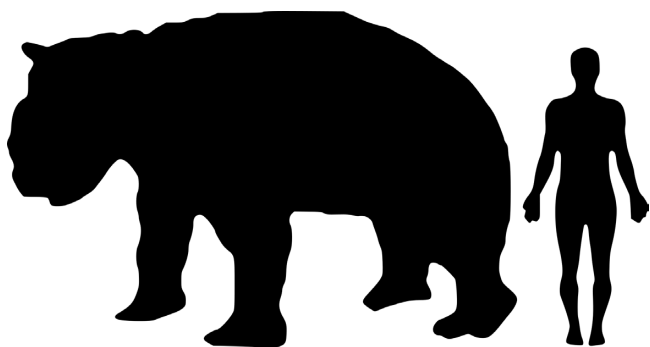
Evidence box 2

Geological epochs are:

Eocene	56–34 million years ago (mya)
Oligocene	34–24 mya
Miocene	24–5 mya
Pliocene	5–1.65 mya
Pleistocene	1.65 million to 10 000 years ago
Holocene	10 000 years ago to the present

Evidence box 3

Diprotodon optatum – a giant wombat – weighed about 2.5 tonnes, and is probably the largest marsupial ever to exist. The common wombat, *Vombatus ursinus*, only reaches a maximum of about 40 kg. The echidnas and kangaroos of today are also considerably smaller than their extinct counterparts.



Evidence box 4

Australia split from Gondwanaland about 40 mya and since then has gradually become more arid, i.e. there is less rainfall.

Evidence box 5

The exact date when Aborigines first arrived in Australia is not known. Most sources indicate that the first humans came to the continent from South-East Asia between 50 000 and 70 000 years ago.

Evidence box 6

The 'blitzkrieg' theory blames Aborigines for the extinction of megafauna. Aborigines hunted and ate them. Bones of megafauna have been found with butchering marks on them. In addition, Aboriginal use of fire destroyed the ecosystems in which the megafauna thrived.

Evidence box 7

Radiocarbon dating can only be used with confidence with samples less than 40 000 years of age. Dating older specimens may only be accurate to about 500 years.

Evidence box 8

Some scientists think that extreme climatic changes, especially drought conditions about 18 000 years ago, were the major factor leading to the megafauna dying out.

Evidence box 9

Commentators have pointed out that water buffaloes are similar in size to megafauna. Efforts to eradicate them in the Northern Territories using more sophisticated weapons than those available to Aborigines have been unsuccessful.

Evidence box 10

Megafauna survived many extremes in temperature and water availability and there are always areas that are favourable to some fauna, so critics of the climatic change model argue as to whether this was a major factor in their extinction.

Interpreting the data

1 Complete the table.

Event or evidence	When did it occur (epoch)?

2 How many times heavier was the diprotodon than the present common wombat?

Analysing the data

3 Summarise the case for and against climate change being responsible for the extinction of Australian megafauna.

4 Summarise the case for and against humans being responsible for the extinction of Australian megafauna.

Questions

5 Scientists have relied heavily on fossil records to build up theories of why Australian megafauna died out. What are the drawbacks of this method?

6 Give the scientific name for diprotodon. _____

7 Give the scientific name for the common wombat. _____

8 What is the difference between extinct and extant?

Data

The type of teeth (dentition) of an animal can give a clue as to its diet. Incisors are for snapping, molars are for grinding, and canines are for tearing flesh. ‘Diprotodontia’ means ‘two front teeth’. Diprotodonts have a pair of large forwardly directed incisors on the lower jaw and three upper incisors. There are four large, simple molars in each jaw.

Making links

9 What was the likely diet of a diprotodont? Justify your answer.

INTRODUCTION

In order to manage and control native and feral animals, agencies have to know where a particular species tends to be located (distribution) and how many individuals there are in a particular area (abundance). Because these animals are often nocturnal, living in remote areas or wary of humans, a wide range of sampling techniques are used. These include collecting faecal samples (scats) and using sand plots.

Aim

To gain a greater understanding of the various techniques used to gather data on the abundance and distribution of particular fauna.

Information: Scats

The excreta (scats) of animals can give information on their diet and how it changes with locations and seasons. Predator scats consist mainly of the parts of their prey that can't be digested. The collection of scats is relatively simple, the scats can be stored and analysed when it is convenient, and there is no effect on the wildlife being studied. The shape as well as the contents can help identify the animal being studied. Usually the samples are heated before analysis to remove water and kill any pathogens present. Identifying the contents usually involves microscopic analysis and comparison with known samples. The red fox, *Vulpes vulpes*, is an introduced predator that has contributed to the decline of some native species and has been declared a vermin species. Wild dogs, *Canis lupus familiaris*, pose similar dangers.

Data

The table below shows the number of scats containing hair and bones from various mammals. The data were collected over the period of a year in an area of a national park in NSW. Note that the data are not independent, because a scat can contain the remains of more than one animal.

Animal	Black rat	Brushtail possum	Common wombat	Grey kangaroo	Greater glider	Rabbit	Sugar glider	Swamp wallaby
Feral dog	0	0	18	47	19	25	16	220
Fox	25	23	0	0	24	138	22	141

Analysing the data

1 Using data from the table, draw a column graph showing the relative composition of scats from feral dogs and foxes.



2 Discuss any patterns in the predators and their prey.

Questions

3 What are scats?

17 – Gathering data on distribution and abundance of animals continued

- 4

Why are scats heated before they are analysed?
- 5

Describe the advantages and drawbacks of using scats to study the diets of predators.
- 6

What is a ‘vermin’ species’?
- 7

A grazier in the area studied insisted that feral dogs had been attacking his sheep. However there were no traces of sheep in the scats of the feral dogs or foxes studied. Give possible reasons for this.

Information: Sand plots

Looking at animal tracks has been used for many years, but not all surfaces give an imprint when an animal steps on them. Sand plots are specially constructed areas of sand or fine soil placed at regular intervals along a track or path used by animals. They are checked at regular intervals. Sand plots are a good way of estimating changes in animal activity and hence an indirect method of monitoring animal populations.

Data

The table below shows the number of tracks attributed to canids – dogs, foxes and dingoes – and other animals recorded at one plot on five consecutive years.

Year	Day	Canids	Other mammals	Marsupials
1	1	1	6	1
	2	1	5	2
	3	1	7	2
2	1	3	4	3
	2	4	5	2
	3	2	2	4
3	1	6	0	0
	2	7	3	0
	3	5	4	3
4	1	4	1	1
	2	1	2	1
	3	1	2	6
5	1	4	1	0
	2	3	2	0
	3	5	0	0

Analysing the data

- 8

Using data from the table, calculate the average daily number of tracks for the three groups of animals over the time studied. Show your data in table form.

17 – Gathering data on distribution and abundance of animals continued

9 What conclusions, if any, can be drawn from your results? State your reasons.

10 Discuss any limitations in the method used and how it could be improved.

Information: Global positioning systems

For many years the only way to get ‘real-time’ data on the position of a wild animal was to fit it with a radio collar. This transmits a UHF signal and has to be manually tracked using a portable aerial. This technique is labour intensive, requires the tracker to follow the studied animal and usually only one animal can be monitored at a time.

The introduction of microelectronics and GPS systems has given researchers greater flexibility and more information. An animal is trapped and fitted with a GPS collar. Data, such as latitude, longitude and time, are then recorded at regular intervals. Collars have to be light, robust and have a good battery life. Many are designed to release themselves from the animal after a period of time.

Dingoes, *Canis lupus dingo*, have been implicated in the killing of sheep.

Data

The tables below show the movements of a rural dingo and an urban dingo recorded on two areas of similar size, e.g. 15 km by 15 km grids, over two days at similar times of the year. X and Y are the grid coordinates.

Rural

Elapsed time (h)	0	2	4	6	8	10	12	14	16	18	20	22
X	13	11	*	8	7	6	6	6	6	6	3	*
Y	14	14	*	15	14	10	10	10	10	10	8	*
Elapsed time (h)	24	26	28	30	32	34	36	38	40	42	44	46
X	*	*	2	3	3	3	8	8	11	11	12	12
Y	*	*	5	4	4	4	4	4	2	3	3	2

Urban

Elapsed time (h)	0	2	4	6	8	10	12	14	16	18	20	22
X	3	4	5	5	5	5	6	8	9	10	11	11
Y	11	10	8	8	8	7	2	4	6	8	8	9
Elapsed time (h)	24	26	28	30	32	34	36	38	40	42	44	46
X	13	13	13	13	13	8	7	7	6	6	6	7
Y	10	10	10	10	10	11	7	7	10	12	12	11

* = not on grid

Questions

11 What advantages do GPS collars have over conventional radio collars?

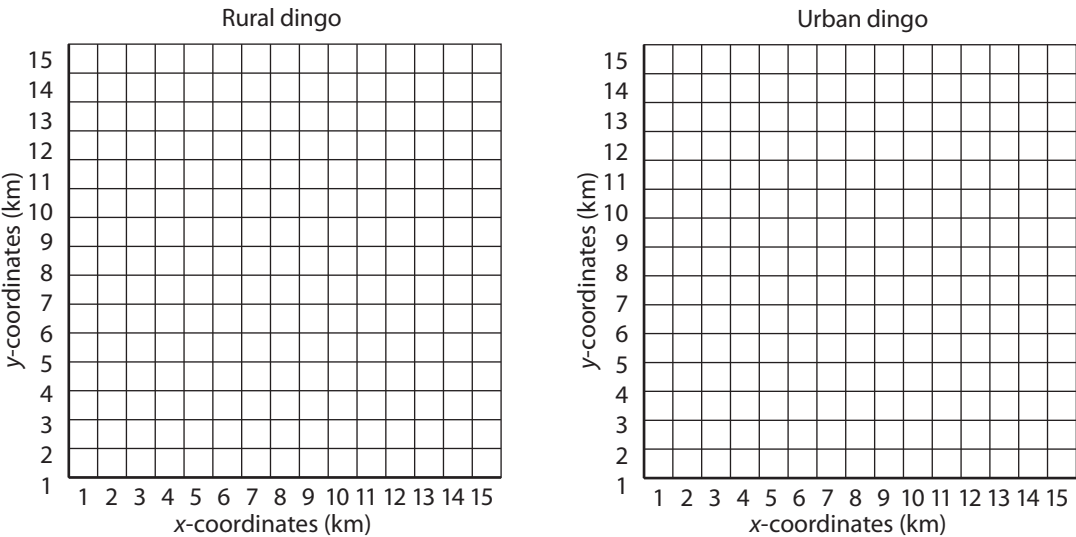
12 Why might it be a good idea to have collars that automatically detach from the animal after a period of time?

13 What is meant by ‘implicated in the killing of sheep’?

Analysing the data

14 A previous study had found an average population density of 18 dingoes per 100 km² in the rural area. Calculate the number of dingoes likely to be present in the rural area. Show working.

15 Using data from the tables on the previous page, plot the positions of the dingoes.



16 Compare the movements of the two groups of dingoes.

17 Discuss any limitations in the method used and suggest how it could be improved.

Information

Dingoes have been seen to take sheep and this has been confirmed by a study of their scats (faeces). Many methods have been tried to stop this happening. These have included physical barriers such as fences, but these are expensive to erect and maintain. Poison bait (e.g. 1080, sodium fluoracetate) is relatively cheap and effective, but care must be taken to ensure that other animals aren’t poisoned. Shooting has been tried and bounties offered, but although this method is specific it too is labour intensive.

Treating the data

18 In a table, set out the advantages and disadvantages of the methods of dingo control described above. Include at least one possibility not mentioned in the text.

Neurones and nerves

Neurones (neurons) are nerve cells and transmit signals by electrochemical changes in their membranes. They cannot be seen without specialised techniques and equipment, e.g. staining and using electron microscopes. Nerves are bundles of neurones.

Aim

To improve knowledge of how neurones work.

Information

A neurone consists of dendrites, a cell body, an axon and the synapse. Nerve cells usually have several dendrites (branches) which increase in thickness as they get nearer to the cell body. Nerve signals travel from the dendrites to the cell body and then to axons. Axons are long and thin and often coated with an insulating layer of myelin. Nerve signals travel away from the cell body. A synapse is the junction between the axon of one neurone and the dendrite of the next. When an electrical signal reaches the end of the axon, a chemical (neurotransmitter) is released and travels across the junction. The dendrite that receives the neurotransmitter then sends an impulse to the cell body and the axon. Axonal nerve signals are called action potentials.

Nerve cells have to be stimulated before they can send off an impulse. Neurones have an 'all or nothing' response. Not all stimuli cause a response, i.e. an action potential; stimuli have to reach a 'threshold' or minimum value before an action potential is generated.

Interpreting the data

- I** Figure 1 shows two linked neurones. Label the axon, cell body, dendrite, myelin and synapse.

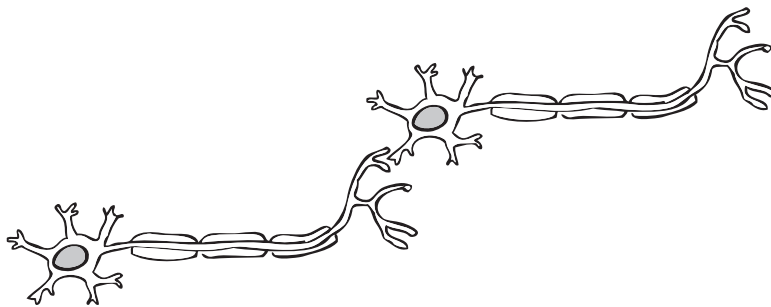


Figure 1

- 2** Figure 2 shows a typical action potential. Explain what is happening.

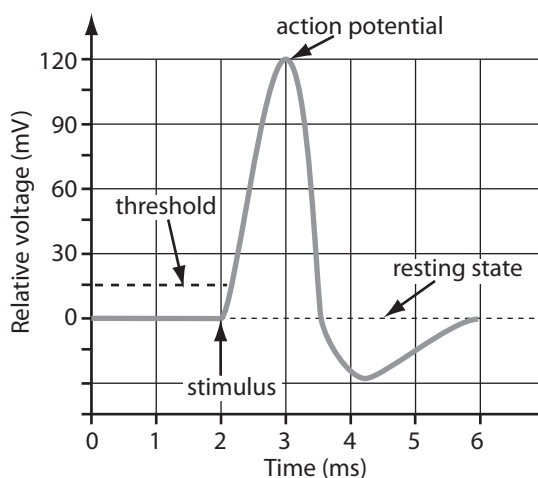
[illegible]

Figure 2

INTRODUCTION

The human brain is part of the central nervous system and controls and coordinates our conscious and unconscious actions.

Information

The three major regions of the brain are the cerebrum, the cerebellum and the medulla oblongata. The cerebrum is the largest part of the brain and also the newest, evolutionarily. It is the front and upper part of the brain and is divided into two hemispheres. Its surface is called the cerebral cortex. The cerebrum is made up of four lobes (subdivisions). These are the frontal lobe (speech), the parietal lobe (general sensory area), the occipital lobe (vision) and the temporal lobe (auditory). The medulla oblongata links the brain and the spinal cord and controls many autonomic functions such as heart beat and breathing. Figure 1 shows the left exterior of the human brain.

As we have evolved, the size of our brains has increased. A human brain is approximately three times the size of the brain of a mammal of equivalent mass, and has approximately 50 to 100 billion neurones.

Aim

To outline the major areas of the brain and their functions.

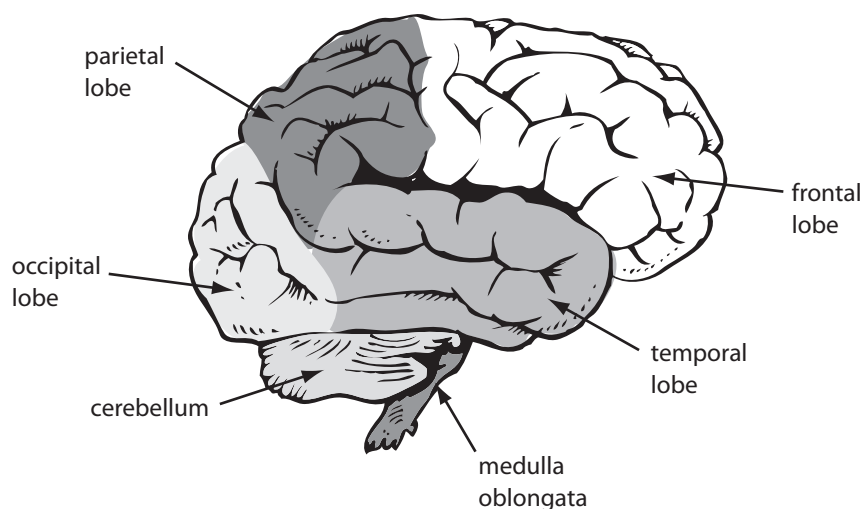


Figure 1

Treating the data

- 1 Write the function of each of the various parts of the brain next to its name in Figure 1.

Questions

- 2 What is meant by an autonomic function? _____
- 3 What is a lobe? _____
- 4 What are the likely results of humans having a comparatively much larger brain than similar-sized mammals?

- 5 What is another name for a neurone? _____
- 6 Why might a large animal with a large brain show no more intelligence than a smaller animal with a smaller brain?

Research

- 7 What is meant by cranial capacity?

1 Vaccination and the prevention of disease

Common name	Scientific name	Symptoms
Chicken pox	Varicella	Itchy rash and red spots
Whooping cough	Pertussis	Similar to common cold, coughing
Polio	Poliomyelitis	Flu, paralysis
Smallpox	Variola major	Headache, fever, small red spots, blisters

- 2 For: Low risk, less chance of anybody catching disease, saves lives.
Against: Can cause illness, chances of catching a disease are already low.
- 3 Someone from abroad carrying the disease entered the country.
- 4 It is the study of the distribution and frequency of diseases.
- 5 It is not recommended if they are too young or have a compromised immune system.
- 6 Immunoglobins
- 7 Most of the community has been vaccinated and has immunity so there is not a big enough pool of people to sustain or spread the disease.

2 Classification (taxonomy)

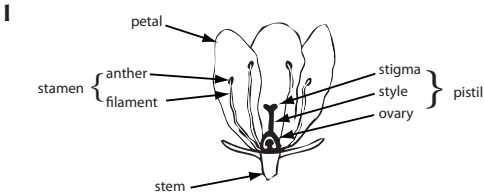
- 1 Animal & animalia; spinal cord & chordata; suckles young & mammalia; opposable thumb (or big toe) & primates; bipedal & hominidae; human & homo; wise & sapiens
- 2 Only one type of animal can exist as a species, more types can be part of a genus; there are even more types in a family and so on.
- 3 To ensure they can be specific about the organism they are describing and that each animal has a unique name
- 4 Two words are used and both are written in italics. The first refers to the genus (using a capital letter) and the second describes the species.
- 5 e.g. Kind Phillip Came Over For Grape Soda

Animal	Class	Dorsal nerve cord	Jaws and spinal column	Lungs	Claws or nails	Feathers	Fur (hair) and mammary glands	Placenta
Chimp	Mammal	✓	✓	✓	✓		✓	✓
Frog	Amphibian	✓	✓	✓				
Hagfish	Agnatha	✓						
Kangaroo	Mammal (marsupial*)	✓	✓	✓	✓			
Lizard	Reptile	✓	✓	✓	✓			
Pigeon	Bird	✓	✓	✓	✓	✓		
Tuna	Fish	✓	✓					

*Marsupials are a subclass within the mammal class. (Taxonomists often differ on classification.)

- 7 e.g. Brain, paired limbs, able to move on land, young born out of water, ability to fly, give birth to live young, opposable thumb/big toe
- 8 Humans would appear after the chimp. They occur at a later stage in evolution.
- 9 Humans have a permanent upright stance, larger cranial capacity (size of brain is over 1000 mL for humans, about 400 mL for chimp), and almost total lack of insulating fur on body.
- 10 Genetic information (e.g. DNA sequences), how the animals behave (e.g. social structure), means of locomotion (fly, walk, swim)

3 Angiosperms



Name	Description
Anther	Male structure
Filament	Female gamete
Flower	Has potential to form new plant
Ovary	Thin stalk
Ovule	Location of female gametes
Pistil	Location of male gametes
Pollen	Female structure
Seed	Tube linking stigma to ovary
Stamen	Sex organ of plant
Style	Male gamete

3

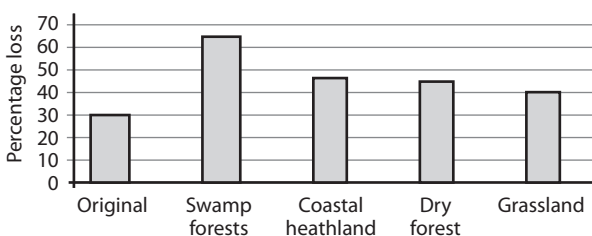
Type of pollination	Advantages	Disadvantages
Wind	Only need unobtrusive flowers Better for dense growths Better for plants reaching maturity at same time	Need large amounts of pollen
Animal	Less pollen More efficient Pollen moved over greater distance	Have to produce nectar

- 4 *Plantago gaudichaudi* is wind-pollinated – the anthers are projecting from the plant. *Banksia ericifolia* is insect pollinated – the anthers are not exposed.
- 5 Seeds, nectar
- 6 Smell, bright flowers
- 7 a Basically most plants that have the properties described in (6) e.g. rose, tulip
b e.g. Bird of paradise (*Strelitzia reginae*), waratahs
c e.g. *Banksia*, Proteaceae
- 8 Increases their ability to survive environmental changes or disease
- 9 e.g. Environmental conditions such as lack of water or outbreak of disease

4 Global warming – possible impacts on the Pacific

- 1 15°C, 27°C
- 2 20°C, 32°C
- 3 It assumes current trends will continue at the same rate.
- 4 Humans could change their behaviour and use of energy.
- 5 More likely to have scientific institutions that can gather accurate data over a long period.
- 6 Yes. All areas of the Earth are linked by atmosphere and oceans.
- 7 Sea levels are fluctuating and seem to have been rising in recent years.
- 8 The longer term results should give an accurate trend. Using only recent data can emphasise short-term fluctuations.
- 9 Include flooding of coastal areas, less land available, salinity reaching inland
- 10 Island B. A small rise in sea level would cover a greater area.
- 11 Small and very remote
- 12 They would move to somewhere not as likely to be rendered uninhabitable by the rise in sea level.

5 Biodiversity

- 1 
- 2 $7 \times 10\,500 = 73\,500$ ha
- 3 There is a loss of habitat/food for these organisms.
- 4 a Varied
b Group of similar organisms that can breed and produce fertile offspring
c Wide range of species
- 5 The strategies can be legal and involve penalties (1, 4), forward planning and permission (3), research (6), helping to breed endangered/scarse species (5) and helping to restore/improve habitat (2).
- 6 It could include publicity to encourage the public to support biodiversity, set aside land, sponsorship of projects.

6 Invasive plants and their control

- 1 No natural predators, conditions (nutrient levels and water temperature) suit the plant
- 2 Fishing, swimming, boating, water skiing

3

Control method	Advantage	Disadvantage
Declared pest	Legal, compulsory	Difficult to enforce
Mechanical means	Low technology	Labour intensive, not suitable for large growths
Herbicides	Effective	Pollution of drinking water
Biological controls	Little human intervention required	Can become a nuisance

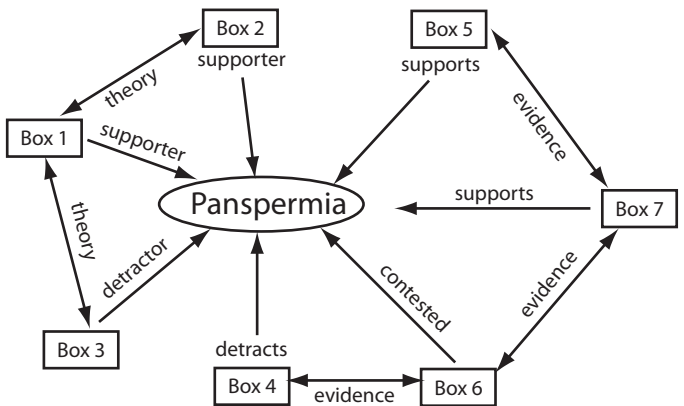
- 4 80 squares; *S. molesta* covered a total of 43 squares (approx). Cover = $(43/80) \times 100 = 54\%$
- 5 Only approximate. Have to estimate by adding up fractions of squares. More accurate if using smaller squares and/or more squares.

- 6 Colder weather due to time of year
- 7 Warmer weather
- 8 Timing and temperature were more suitable. Weevils were able to eat buds of growing plants.

7 Exogenesis and the origin of life on Earth

- 1 1, 5 and 7
- 2 3, 4 and 6
- 3 Has merit. Life must have started somewhere at some time.
- 4 a Not derived from living things
- b Between stars
- c Experiment in which the variables/conditions are controlled and against which others are judged and variables changed.
- d Sum of all chemical reactions in an organism
- e From outside of Earth
- f Can survive extreme conditions
- g Presence of unwanted substance
- 5 e.g. There were traces in the Orgueil meteorite (1966) that closely resemble fossilised magnetotactic bacteria of the species *Rhodopseudomonas rutilis*. Two Italian scientists from the University of Naples claim to have discovered bacteria in a meteorite that is over 4.5 billion years old (2001).
- 6 No planets that would support life have yet been discovered. Fossil evidence points to life having evolved on Earth.
- 7 Geogenesis is the theory that life originated on Earth. Complex chemicals formed which were the precursors of primitive life. This life evolved over billions of years to the variety of organisms in existence today.

8 e.g.



8 The platypus – a case study

- 1 5 × 7 cm = 35 cm
- 2 Order: Monotremata, genus: *Ornithorhynchus*, species: *anatinus*
- 3 Mammal: endotherm, fur, tail like a beaver, feet like an otter, mammary glands. Other classes: lower body temperature, lays eggs, bill like that of a duck, venom in spurs
- 4 Person who classifies organisms
- 5 Approximately 75 million years
- 6 130 mya
- 7 Contradicts. Split around 150 mya, when primitive mammals separated into prototherians and therians.

8	Mammal	Lay eggs	Have a pouch	Suckle young	Location of species	Endothermic or ectothermic	Covered in skin and fur
	Monotremes	✓	✓	✓	Australia	Endothermic	✓
	Marsupials	✓	✓	✓	Australia	Endothermic	✓
	Placentals		✓	✓	World wide	Endothermic	✓

- 9 Plate tectonics
- 10 a Coat of a mammal
- b Can exist only in a particular environment
- c Tends to be active during the hours of darkness
- d Length of time between fertilisation and birth

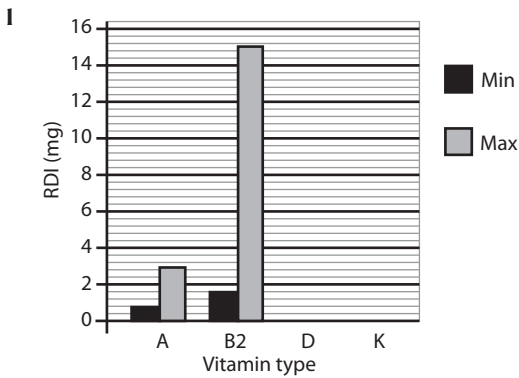
9 Water quality

- 1 (n/a = not available)

State	2002			2004			2009		
	Good	Moderate	Poor	Good	Moderate	Poor	Good	Moderate	Poor
A	n/a	n/a	n/a	55	32	13	66	27	7
B	n/a	n/a	n/a	78	20	2	75	23	2
C	75	23	2	76	24	0	61	34	5

- 2 For state A, water quality has improved over dates studied. State B water quality has marginally decreased. The water quality of state C deteriorated between 2004 and 2009.
- 3 If greater lengths of waterways were studied, they could have turned up some waterways previously not investigated that were of poorer quality.
- 4 $[(13 \times 10) + (36 \times 10) + (15 \times 8) + (4 \times 8) + ((17 \times 8) + (3 \times 6) + (4 \times 6) + (21 \times 4) + (22 \times 2))] \div 10 = 94.8$
- 5 Excellent
- 6 Food supply, breeding rate of organisms, human intervention
- 7 Micro-invertebrates need a microscope to be seen.
- 8 e.g. Dissolved oxygen, dissolved solids, pH, turbidity

10 Vitamins



- 2 Values of these are very much higher than others, which would have made scale very difficult to draw.
- 3 Vitamin E

4

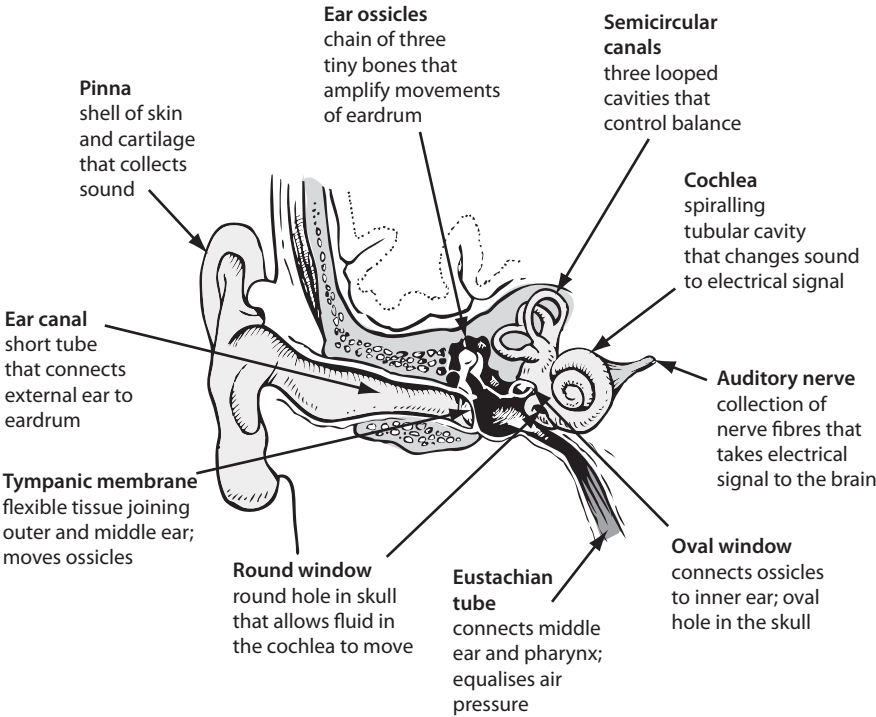
Solubility	Advantage	Disadvantage
Water	Hard to overdose, as body gets rid of excess in urine	Need to take regularly, body doesn't store
Fat	Can be taken at irregular intervals because stored by the body	If too much taken, it is stored in the body and not excreted

5

Vitamin	% RDI (min)	% RDI (max)
A	71	17
B2	40	4
C	134	5.5
D	0	0
E	5333	80
K	0	0

- 6 There is less than the minimum RDI for vitamins A, B2, D & K. Vitamin E is in excess of the minimum RDI, but less than the maximum RDI.
- 7 No. If the rest of the person's diet contains enough of the required vitamins.

11 Structure of the ear



12 Hypotheses and variables

1	Hypothesis	Variables		
		Independent	Dependent	Controlled
	Whiteflies are attracted to the colour yellow.	Colour of plastic	Numbers of whiteflies trapped	• Size and shape of strip • Amount of oil

2 Repeat. Try strips in a different place. Use a different coating.

3	Hypothesis	Variables		
		Independent	Dependent	Controlled
	Junket tablets set milk quickest at a specific temperature.	Temperature of water bath and milk	Time to set	• Volume of milk • Amount of junket

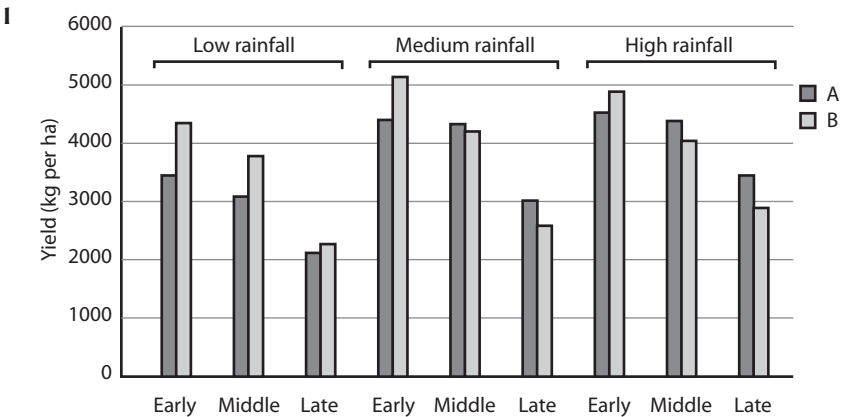
- 4 Repeat at intermediate temperatures, e.g. 15°C, 25°C, 35°C and 45°C.
- 5 Repeat experiment but adjust the pH of milk (add acid or base). The independent variable is the pH of the milk. The dependent variable is the time taken by milk to set. Controlled variables are the volume of milk, temperature of milk and amount of junket.

13 Crocodiles – a case of arrested development?

1	Saltwater crocodile	Freshwater crocodile
	Found in salt water and fresh water	Prefers fresh water
	More aggressive	Carnivorous
	Carnivorous	Breeds in dry season
	Larger	
	Broader snout	
	Breeds in wet season	

- 2 It is unlikely that any (more aggressive) saltwater crocodiles are present.
- 3 Human = 1.6 m = 31 mm, hence 1 mm = 0.052 m X = 90 mm = 4.65 m, Y = 43 mm = 2.22 m
- 4 X is a saltwater crocodile (larger) and Y is a freshwater crocodile (smaller).
- 5 Length of track of identified crocodile is 37 mm. This crocodile is 2.9 m in length. Unidentified track is 26 mm. Hence length of unidentified crocodile is $(26 \div 37) \times 2.9 = 2.0$ m. Assume that size of track is proportional to overall size.
- 6 The deeper the tracks, the heavier the crocodile; the larger the tracks the longer the crocodile.
- 7 a, e, f, i, l, p, r, s, t. (Some answers vary in degree of importance depending on specific reptiles.)
- 8 b, d, g, h, j, k, q, r, s 9 b, d, g, h, j, k, l, m, n, o, r, s, t

14 Crop yields



- (Note: y-axis is yield in kg per hectare; x-axis shows A and B in low (L), medium (M) and high (H) rainfalls with early, middle and late sowings.)
- 2 In general, the earlier the sowing the greater the yield for both varieties. In all areas, maximum yield is obtained by planting varieties early. If you want a maximum yield of A, plant early in a high rainfall area. If you want a maximum yield of B, plant early in a medium rainfall area.
- 3 300 ppm
- 4 a Increase is $(5.2 - 4.4) / 4.4 \times 100 = 18.2\%$ b $(5.8 - 4.4) / 4.4 \times 100 = 31.8\%$ c $(6.0 - 4.4) / 4.4 \times 100 = 36.4\%$
- 5 Plant has reached maximum rate of uptake of carbon dioxide or photosynthesis.
- 6 Carbon dioxide plus water (in the presence of light) forms glucose and oxygen.
- $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

15 Chronic heart failure (CHF)

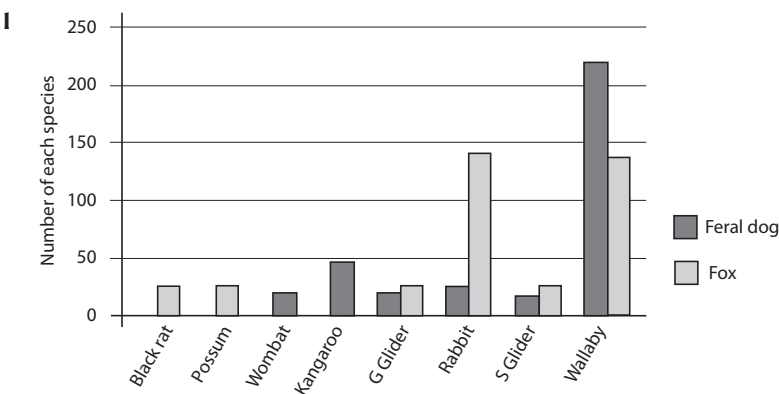
- 1 Alcohol abuse, obesity, smoking, diet
- 2 Shortness of breath, kidney function is impaired, congestion of lungs, less efficient intestines, death
- 3 Monitor weight, fluid intake, caffeine intake, manage stress
- 4 Slow the loss of heart-pumping ability
- 5 Improve strength of heart’s contractions
- 6 Blood flow is routed around the blocked blood vessel.
- 7 $(21\,500\,000)/1000 \times 1.2 = 25\,800$
- 8 Increased awareness of problems, adopting healthier lifestyle, better drugs, improved surgical procedures
- 9 LVADs assist heart; TAHs are complete artificial replacements.
- 10 24%
- 11 There is a shortage of suitable donors.

16 Australian megafauna

1	Event or evidence	When did it occur (epoch)?
	Megafauna flourished and then became extinct	Pleistocene
	Australia split from Gondwanaland	Eocene
	Aborigines arrived in Australia	Pleistocene
	Extreme climate changes	Pleistocene

- 2 $2500/40 = 62.5$
- 3 For: Climate change can alter conditions severely and make conditions difficult for some species. There were climate changes at approximately the time of megafauna dying out.
Against: Megafauna survived similar, earlier changes. Conditions always suit some animals.
- 4 For: Aborigines hunted and ate them. Fire may have destroyed their habitat.
Against: Aboriginal weapons were not capable of killing large numbers of big animals.
- 5 Fossil records are often incomplete and sometimes difficult to date with accuracy.
- 6 *Diprotodon optatum*
- 7 *Vombatus ursinus*
- 8 Extinct: no longer exists; extant: still in existence
- 9 Vegetation. No canines for eating flesh. Incisors and molars best for dealing with vegetation.

17 Gathering data on distribution and abundance of animals

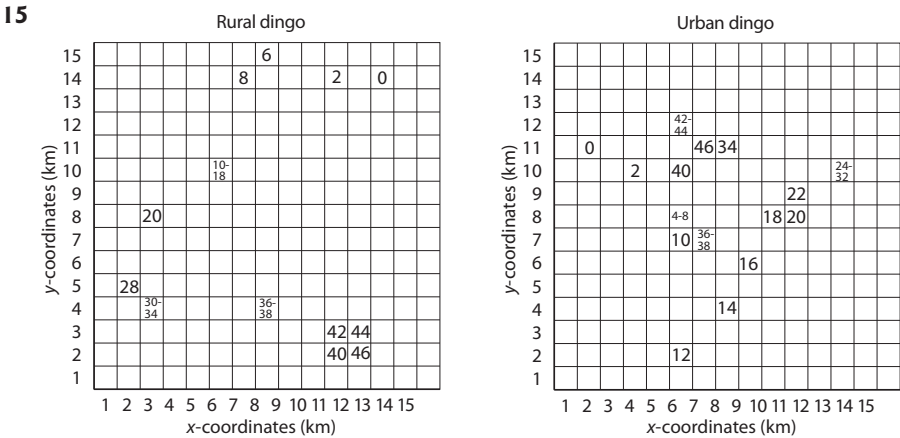


- 2 Fox seems to prefer smaller animals. Wallaby is the major prey of feral dogs and foxes.
- 3 Scats are the excreta of animals.
- 4 They are heated to remove water and kill any pathogens.
- 5 Advantages: Collection is simple, they can be stored and analysed when convenient and it doesn’t affect animal being studied.
Disadvantages: Needs microscopic analysis and comparison. Only contains indigestible parts – completely digested animals can’t be identified.
- 6 Destructive, annoying, or injurious to health; often present in large numbers
- 7 Could have attacked sheep without eating, or another group may have been responsible.

8

Year	Canids	Other mammals	Marsupials
1	1	6	1.7
2	3	3.7	3
3	6	2.3	1
4	2	1.7	2.7
5	4	1	0
Overall	3.2	2.9	1.7

- 9 Overall canids are most common. Numbers fluctuate, but numbers of other mammals seems to be decreasing.
- 10 It only records animals crossing that area. May be avoided by some animals. Same animal may cross many times. Need to record over greater time and greater areas. Might be better to use cameras to record.
- 11 Less labour intensive, greater flexibility and provide more information
- 12 So animal is not hurt by collar.
- 13 Evidence suggests that they killed sheep.
- 14 Area is $15 \times 15 = 225 \text{ km}^2$; $(225/100) \times 18 = 40.5$

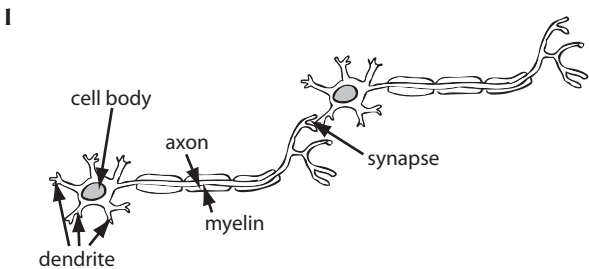


- 16 Rural dingo travels further afield. Urban dingo has more time in specific locations.
- 17 Positions of dingoes at times were not known – off grid. Greater grid needed.

18

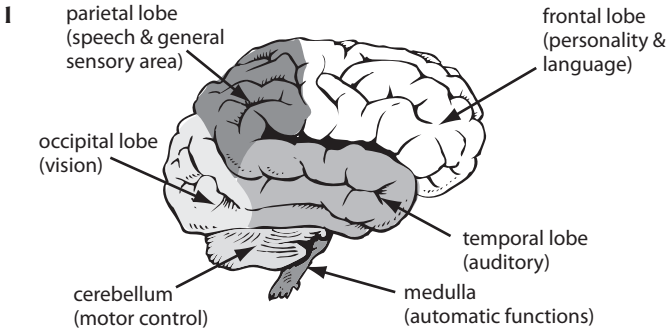
Method	Advantages	Disadvantages
Physical barrier	Low level of technology	Expensive to erect and maintain; restricts all animals
Poison bait	Cheap and effective	Can poison other animals
Shooting	Specific. Could be carried out by sports hunters	Labour intensive

18 Neurones and nerves



- 2 Nerve cell is resting. A stimulus (of the required minimum value) occurs and the nerve responds by generating an action potential (voltage) that lasts for about 1.5 ms. After a further 2 ms the nerve returns to its normal resting state.

19 The brain



- 2 One that is carried out without thinking
- 3 It is a subdivision, part of an organ.
- 4 Greater intelligence
- 5 Nerve cell
- 6 A large part of the brain's capacity is taken up with autonomic functions and controlling the body.
- 7 It is the volume of the brain case/brain pan, which is an indication of the size of the brain.

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Reinforcement of learning	✓	A fresh approach to a topic	✓
Extension work for advanced students	✓	A way to target student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use the lessons to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Interpreting data in Senior Biology

This senior secondary Biology resource includes self-contained units of work that will refine students’ ability to read, describe, analyse, manipulate and interpret data. Students create diagrams, complete tables and form thoughtful written responses about topics such as water, the desalination process, crop yields, global warming and biodiversity. Each unit follows a stepped process that will ensure that students develop appropriate science method skills. Extended answers have been included to assist with marking and to promote a complete understanding of the topic.

Topics include

- Vaccination and the prevention of disease

Angiosperms

Global warming – possible impacts on the Pacific

Biodiversity

Invasive plants and their control

Exogenesis and the origin of life on Earth
- The platypus – a case study

Water quality

Crocodiles – a case of arrested development?

Chronic heart failure

Australian megafauna

Neurones and nerves

Associated titles in the Instant Lessons series

- Interpreting data in Senior Chemistry

• Interpreting data in Senior Physics

• How Science Works
 - Biology
 - Chemistry
 - Physics
- Understanding Biology
 - Ecosystems and living things
 - Biotechnology and disease
 - Human and plant life

• Practical Experiments in Biology (Books 1,2,3,4)

Visit the Blake Education website or email info@blake.com.au for more information about the Instant Lessons series and ordering advice.